

# Beyond the Jack-of-All-Trades Hypothesis: The Importance of Business and Specialized Skills for Self-Employment Entry and Success

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## Abstract

What types of skills does a worker need to start and run a business? A leading explanation, Lazear's (2004; 2005) jack-of-all-trades hypothesis, holds that workers with balanced skills are more likely to become entrepreneurs. Empirical studies support the hypothesis, finding that workers who have held more occupations or employers, a common proxy for skill balance, are more likely to be self-employed, with all forms of self-employment pooled together. I revisit the hypothesis in three ways. First, I argue that what prepares a worker for business ownership is not balance across all skills but two types of skills: business skills and specialized skills relevant to the business. Second, I distinguish between basic and advanced self-employment, where advanced businesses are incorporated or more capital-intensive at entry. Third, I build a direct measure of skill balance and compare it with the conventional occupation count. I present a static choice model in which entry depends on the returns to skills in self-employment relative to those in wage employment, whereas success (revenue and earnings) depends on the returns to skills within self-employment. Using nearly forty years of NLSY79 work histories linked to O\*NET, I find that business skills raise entry into advanced but lower entry into basic self-employment, while specialized skills lower entry into both. After entry, business skills predict success in advanced self-employment and specialized skills in basic self-employment. These results have three implications for the JOAT hypothesis. Skill balance, measured directly, predicts neither entry nor success; pooling basic and advanced self-employment masks how the roles of business and specialized skills differ; and the occupation count predicts entry into basic self-employment partly because it reflects involuntary job losses rather than accumulated skills. Owners of advanced businesses are jacks of business trades rather than jacks of all trades.

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# 1 Introduction

What kind of work experience prepares someone to start a business? The best-known answer comes from Lazear’s (2004; 2005) jack-of-all-trades (JOAT) hypothesis. Because entrepreneurs must handle many functions at once, such as marketing, finance, and operations, workers with balanced skills have a comparative advantage in entrepreneurship, while specialization ties workers to wage employment. An active literature finds empirical evidence in support of this hypothesis (Wagner, 2003, 2006; Stuetzer, Goethner and Cantner, 2012; Stuetzer, Obschonka and Schmitt-Rodermund, 2013; Bublitx and Noseleit, 2014; Aldén, Hammarstedt and Neuman, 2017). The standard test relates a measure of skill balance to the probability of becoming an entrepreneur. Skill balance is usually proxied by the number of occupations or employers a worker has held, and entrepreneurship is defined as any self-employment and/or business ownership.

I revisit this question by making three contributions. First, I argue that what prepares a worker for business ownership is not balance across all skills, but two types of skills relevant to establishing and operating a business: *business skills*, meaning core business knowledge, managerial skills, and interpersonal skills, and *specialized skills*, meaning expertise directly relevant to the business. This distinction is already present in Lazear’s own example of a founder: “Consider the founder of a new small restaurant. In addition to being a good cook, the founder must be able to obtain funds, hire workers, choose location and decor, obtain food supplies at a reasonable cost, keep books, and market the restaurant” (Lazear, 2004, p. 208). Being a good cook requires expertise specific to the business; the remaining tasks are skills needed to operate a business of any kind. However, in subsequent empirical work, including Lazear’s own test (Lazear, 2005), this distinction is lost: researchers consider skill balance by counting occupations, implicitly giving each occupation equal weight. For example, a worker who has been a truck driver, a chef, and a product manager before opening a restaurant brings three very different types of experience: her years as a chef built the expertise the restaurant needs, her years as a product manager built skills to run it,

and her years as a truck driver built neither, even though each job contributes equally to a simple count of occupations. I therefore shift the question from how balanced a worker’s experience is to which types of skills that experience builds.

Second, I distinguish between two types of self-employment. Most studies pool all self-employment and/or business ownership, so a freelance house cleaner and the founder of an incorporated construction firm fall into the same category, even though they perform different functions and may require different skills. The JOAT hypothesis applies most directly to businesses that are complex enough to require coordinating multiple functions, and pooling different types of self-employment can obscure the roles that different skills play. I classify businesses that are incorporated or start with higher levels of capital as *advanced* self-employment, where owners are more likely to manage functions such as financing, hiring, and marketing in addition to their occupation-specific expertise, and the rest as *basic* self-employment, where the owner’s work may rely primarily on occupation-specific expertise.<sup>1</sup>

Third, I measure skill balance directly. I construct each worker’s *skill portfolio* as the amount of each skill and knowledge element associated with the occupations she has held, weighted by the years she spent in each. I then measure skill balance by how evenly her portfolio is distributed across these elements. This differs from the conventional occupation and employer counts, which are less informative of how her skills are distributed. For instance, five jobs, all in sales, can still produce a narrow set of skills. Count measures may also reflect job mobility rather than skill accumulation, because repeated involuntary job loss can raise them as well. I compare my direct measure with the conventional occupation count, allowing me to test Lazear’s hypothesis on its own terms and to ask whether the conventional proxy captures skill balance or something else.

I formalize these ideas in a static choice model. A worker compares the payoff of wage

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<sup>1</sup>In Lazear’s JOAT hypothesis, the entrepreneur is a business owner, such as the restaurant founder described above, rather than an innovator in the Schumpeterian sense. I use *self-employment* and *business ownership* interchangeably when referring to the broad category of working for one’s own business. My distinction between advanced and basic self-employment is intended to capture differences in the functions performed by owners, with advanced self-employment more closely resembling the businesses envisioned in Lazear’s example.

employment with the payoffs of basic and advanced self-employment, each of which rewards business skills and specialized skills differently. The key mechanism is the opportunity cost of self-employment, the payoff a worker gives up by leaving wage employment. A skill raises this opportunity cost when wage employment rewards it more than self-employment does, and lowers it when self-employment rewards it more. Whether a worker enters self-employment therefore depends on how a skill's return in self-employment compares with its return in wage employment, while how well she does once self-employed depends on the return itself. Under parameter restrictions on the returns to skills that are consistent with the data, the model predicts that business skills raise entry into advanced self-employment and lower entry into basic self-employment, that specialized skills lower entry into self-employment of any type, and that both skills raise payoffs once a worker is self-employed, with business skills mattering more in advanced self-employment. It also implies that a skill can raise the opportunity cost of self-employment, and thereby lowering entry, while still contributing to success once the worker is self-employed.

I test these predictions using near-complete work histories from the 1979 National Longitudinal Survey of Youth (NLSY79), which follows a nationally representative sample of workers over roughly four decades, linked to the skill and knowledge content of each occupation from the Occupational Information Network (O\*NET). I measure business skills and specialized skills by weighting the skill and knowledge content of each occupation by the years a worker spent in it, and skill balance by how evenly her accumulated skills are distributed across skill elements. I estimate how these skills affect entry into each type of self-employment using a competing-risks model and how they predict success once self-employed, measured by business revenue and earnings.

The results support the framework's predictions. Business skills are associated with a 33.6 percent higher probability of entering advanced self-employment and a 14.2 percent lower probability of entering basic self-employment for a half-standard-deviation increase. Specialized skills show the opposite pattern: the same increase is associated with a 13.1

percent lower probability of entering basic self-employment and a 4.5 percent lower probability of entering advanced self-employment. After entry, both business and specialized skills are positively associated with success, but their relative importance differs by type of self-employment. Business skills matter more in advanced self-employment: a half-standard-deviation increase is associated with a 39.4 percent higher revenue and 14.2 to 16.3 percent higher earnings. Specialized skills matter more in basic self-employment, where the same increase is associated with 17.3 percent higher revenue and 14.9 to 19.2 percent higher earnings. Together, these results show that different types of skills play different roles across types of self-employment, and that the same skill can play different roles in entry and success.

Because workers choose the occupations in which they accumulate these skills, the estimated relationships may reflect factors that shape both occupation choices and self-employment. For example, workers with greater wealth, risk tolerance, or preference for autonomy may both select into occupations that build more business skills and be more likely to enter self-employment. I therefore interpret the estimates as conditional associations under a selection-on-observables assumption. The NLSY79 allows me to observe many factors that may shape both skill accumulation and self-employment, including wealth and income, risk tolerance, cognitive ability, personality traits, and family background. For success, the analysis conditions on workers who have already entered self-employment, so the relevant question is how skills are associated with outcomes among those who choose self-employment. As a sensitivity check, I add these rich control variables sequentially and find that the estimated marginal effects of business and specialized skills retain their signs and statistical significance.

My findings have three implications for the JOAT hypothesis. First, what prepares a worker for business ownership is not balance across all skills but business skills and specialized skills relevant to the business. Owners of advanced businesses are jacks of *business* trades rather than jacks of all trades. My direct measure of skill balance is unrelated to entry once business and specialized skills are accounted for, and shows little relationship to success. Second, pooling basic and advanced self-employment hides these patterns. The distinction

matters for the relationship between business skills and entry: because business skills raise entry into advanced self-employment but lower entry into basic self-employment, the effects are offset when the two types are pooled. The distinction also matters for empirical tests of the JOAT hypothesis based on occupation counts. Basic self-employment, the type of self-employment for which the hypothesis fits least well, accounts for much of the relationship between the occupation count and entry when self-employment is pooled. Because these tests do not distinguish between types of self-employment, part of the empirical support for the JOAT hypothesis in the literature may therefore come from basic self-employment. Third, the occupation count, the proxy for skill balance in earlier studies, reflects job churning as well. It predicts entry into basic self-employment only, and about a third of this relationship is explained by involuntary job loss.

These findings also have implications for policies that aim to promote business ownership. For example, the U.S. Small Business Administration (SBA) supports both Small Business Development Centers (SBDCs), which help prospective and existing business owners develop business-related skills, and SCORE, which provides mentoring from experienced business professionals and can connect business owners with mentors with expertise in particular industries or areas. Rather than emphasizing balanced skills, policies could focus on helping prospective business owners build the managerial, financial, and interpersonal skills needed to establish more complex businesses, while industry-specific mentoring could help business owners develop the specialized expertise relevant to their businesses.

## **2 Related Literature**

Lazear's (2004; 2005) JOAT hypothesis formalizes the idea that business owners must be generalists. He argues that because business owners need to handle many tasks, such as marketing, operations, finance, and personnel management, individuals with balanced skills are more likely to enter business ownership than those with deep but narrow expertise. Using

data on Stanford MBA graduates, Lazear (2005) finds that individuals who have worked in more roles and studied more fields are indeed more likely to become business owners.

Researchers have tested the JOAT hypothesis in the context of entry into self-employment using a variety of measures of skill balance. One common approach is to measure skill balance using the number of employers, occupations, industries, or functional areas a worker has accumulated (Wagner, 2003, 2006; Silva, 2007; Åstebro and Thompson, 2011; Stuetzer, Obschonka and Schmitt-Rodermund, 2013; Lechmann and Schnabel, 2014; Chen and Thompson, 2016; Sorgner and Fritsch, 2018). A second approach measures skill balance using the dispersion of ability test scores (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017) or workers' self-reported number of skills and competence across business tasks (Bublitz and Noseleit, 2014; Tegtmeier, Kurczewska and Halberstadt, 2016). Overall, existing studies generally find that individuals with more balanced skills are more likely to enter self-employment.<sup>2</sup> At the same time, some evidence suggests that balanced skills alone may not fully characterize the skills associated with business ownership. Lechmann and Schnabel (2014) and Bublitz and Noseleit (2014) find that business owners possess broader sets of basic-level skills but also require expert-level skills, suggesting that balance in basic-level skills alone is not sufficient for business ownership.

A smaller literature examines outcomes after entry, including self-employment earnings and duration, using the same measures of skill balance described above (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017; Bublitz and Noseleit, 2014; Kurczewska and Mackiewicz, 2020; Koster and Andersson, 2018). Hartog, Van Praag and Van Der Sluis (2010) and Aldén, Hammarstedt and Neuman (2017) find that more balanced skills are associated with higher earnings from self-employment, while Bublitz and Noseleit (2014) finds that the earnings premium is concentrated among individuals with balanced expert-level skills. The literature also examines how balanced skills affect the number

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<sup>2</sup>Silva (2007) finds that the relationship between skill balance and self-employment disappears after controlling for individual fixed effects, suggesting that the observed relationship may partly reflect persistent individual characteristics rather than skill balance itself.

of business gestation activities business owners undertake (Stuetzer, Goethner and Cantner, 2012) and self-employment duration (Koster and Andersson, 2018). However, revenue remains largely unexplored in this literature, despite being a direct measure of business performance and a natural outcome for evaluating the success of business owners.

A related literature suggests that the types of skills workers accumulate matter for business ownership, beyond the balance of skills. In a meta-analysis, Unger et al. (2011) find that human capital related to running a business, such as start-up, industry, and management experience, predicts business success more strongly than education or general work experience. Among UK technology start-ups, Ganotakis (2012) find that commercial, managerial, and same-sector experience predict better performance, while experience in a different sector does not. Within the JOAT literature, Chen and Thompson (2016) find that the relationship between skill balance and self-employment entry is partly explained by having specific types of experience, including finance, business administration, and senior management. These findings suggest that which types of skills workers accumulate may matter more than how balanced their skills are. I build on this idea by using O\*NET skill and knowledge measures to distinguish business skills from specialized skills based on workers' occupation histories (Gathmann and Schönberg, 2010; Lise and Postel-Vinay, 2020). For the interpersonal component of business skills, I use the four O\*NET items that Deming (2017) uses to measure social skills.

Self-employment encompasses businesses that differ substantially in complexity, scale, and capital requirements, which may affect the types of skills that matter. Hurst and Pugsley (2011) show that most businesses are small and have no plans to grow or innovate, arguing that pooling all self-employment and/or businesses can be inappropriate for testing standard theories of business ownership. Evidence from the NLSY79 highlights these differences as well: incorporated business owners perform more nonroutine cognitive tasks and earn more than salaried workers, whereas unincorporated business owners perform more manual tasks and earn less (Levine and Rubinstein, 2017); incorporated businesses also

start with substantially more capital than unincorporated businesses (Levine and Rubinstein, 2018). Moreover, much of the self-employment recorded in the NLSY79 consists of contract or home-based work that respondents do not report as a business (Light and Munk, 2018), and short, low-capital, unincorporated spells are more common among workers with weaker labor market attachment (Humphries, 2017). These suggest that the skills relevant for entry and success in self-employment may depend on the type of self-employment, but the JOAT literature has largely not distinguished among these types.<sup>3</sup> I therefore distinguish types of self-employment using both incorporation status and start-up capital. Incorporation status is more available in existing data, but it may also reflect factors such as tax or liability considerations rather than the underlying characteristics of the business. Start-up capital provides an additional dimension for distinguishing businesses by their scale and capital requirements, although my results are robust to using incorporation status alone.

Finally, the occupation and employer counts commonly used in the JOAT literature are measures of job mobility that are interpreted as proxies for skill balance. This interpretation implicitly assumes that moving across occupations or employers leads workers to accumulate a more balanced set of skills. Job mobility may reflect productive job shopping (Neal, 1999), but it may also reflect poor performance (Groes, Kircher and Manovskii, 2015). Moreover, Chen and Thompson (2016) find that frequent job changes, together with past unemployment, predict entry into self-employment, while Poschke (2013) find that workers with poor labor market prospects tend to start small, short-lived businesses. Some studies directly measure skill balance using the dispersion of ability test scores (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017), but these measures capture skills or abilities observed before labor market entry rather than those accumulated through work. I construct a direct measure of skill balance based on the distribution of skills accumulated over workers' careers and compare it with the conventional occupation-count measure.

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<sup>3</sup>Lechmann and Schnabel (2014) is an exception: they distinguish between solo self-employment and self-employment with employees and find that the positive relationship between the number of occupations and self-employment holds only among the solo self-employed.

### 3 Conceptual Framework

In this section, I set up a static choice model to show how given stocks of business and specialized skills affect a worker's choice between wage employment and two types of self-employment, and how they affect her payoff once she is self-employed. I begin with a brief overview of Lazear's model, which my framework builds on.

#### 3.1 Lazear's Jack-of-All-Trades Model

In Lazear (2004, 2005), a worker has two arbitrary skills,  $x_1$  and  $x_2$ , and chooses between working as a specialist in wage employment and becoming a business owner. A specialist is paid for her stronger skill, so her payoff is  $\max(x_1, x_2)$ . A business owner must perform many tasks herself, so her payoff is limited by her weaker skill. Her payoff is  $\lambda \min(x_1, x_2)$ , where  $\lambda$  is the market value of a business owner's talent. She becomes a business owner when  $\lambda \min(x_1, x_2) > \max(x_1, x_2)$ , which requires her two skills to be at similar levels. Workers with balanced skills therefore become business owners, and workers with one strong skill remain specialists in wage employment.

My framework keeps the basic structure of Lazear's model: a worker with given skills compares the payoff of wage employment with the payoff of business ownership. It differs in four ways. First, the two skills are not arbitrary. They are business skills and specialized skills, the two types of skills that running a business likely requires. Second, I distinguish basic from advanced self-employment, because the skills a business requires depend on its type. Third, the payoff in each employment option depends on both skills, but the return to each skill varies across option. In Lazear's model, specialists are paid only for their stronger skill, whereas business owners are paid only for their weaker skill, making the balance between skills central to the choice. In my framework, each skill contributes to the payoff in every option, but at different rates. As a consequence, a skill can raise the payoff from self-employment while reducing the likelihood of entry if it raises the payoff from wage

employment even more. This distinction separates the determinants of entry from those of self-employment payoffs: entry depends on the relative returns to a skill across options, whereas the self-employment payoff depends on its return within self-employment. This cannot occur in Lazear’s model, where an increase in the weaker skill raises the payoff from business ownership relative to wage employment, making entry more likely. Fourth, I add tastes, which capture all factors other than skills that affect the choice. In Lazear’s model, skills alone determine who becomes a business owner. With tastes, workers with the same skills can make different choices, and the role of skills is isolated in the opportunity cost of leaving wage employment.

## 3.2 Setup

A worker has given stocks of two skills accumulated through prior work experience: business skills,  $BX$ , and specialized skills,  $SX$ .<sup>4</sup> She chooses one of three employment options,  $j \in \{W, B, A\}$ : wage employment ( $W$ ), basic self-employment ( $B$ ), or advanced self-employment ( $A$ ).<sup>5</sup>

Each option pays a return on the worker’s skills. Let  $r_j^{BX}$  and  $r_j^{SX}$  denote the returns to business and specialized skills in option  $j$ . I normalize both returns in wage employment to one, so that the returns in self-employment are measured relative to wage employment. The

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<sup>4</sup>Because the goal of the framework is to show how given skills affect the choice, I do not model how workers accumulate them. Relaxing this assumption is an area for future research. For example, an occupational choice or lifetime skill investment model could be extended to include self-employment as an option, so that workers decide how much of each skill to accumulate while knowing that they may later start a business.

<sup>5</sup>I leave out nonemployment for simplicity. In the data, the alternative to self-employment pools wage employment and nonemployment (Section 4.2). Nonemployment accounts for about 26 percent of the person-years at risk of entry. In practice, workers often take a break between jobs, for example to look for work or to prepare a business, so a period without work is often part of the move from one type of employment to another.

payoffs are additive in the two skills as follows:

$$y_W = BX + SX, \quad (1)$$

$$y_B = r_B^{BX} BX + r_B^{SX} SX, \quad (2)$$

$$y_A = r_A^{BX} BX + r_A^{SX} SX. \quad (3)$$

Skills are not the only factors that determine whether a worker enters self-employment. Workers also differ in wealth, risk tolerance, personality traits, preference for autonomy, business aspirations, and many other characteristics that make self-employment more or less attractive to them. I summarize all of these factors in two parameters,  $\tau_B$  and  $\tau_A$ , which measure how much the worker values basic and advanced self-employment relative to wage employment, and I refer to them as *tastes*. The taste for wage employment is normalized to  $\tau_W = 0$ . Tastes vary across workers with joint density  $g(\tau_B, \tau_A)$ , and  $\tau_B$  and  $\tau_A$  can be correlated. For example, a worker who values autonomy is likely to value both types of self-employment. Tastes change how attractive an option is, but they do not change the returns to skills in that option.

Because I take both skills and tastes as given and model only how they affect the choice, tastes do not influence how much of each skill a worker has in this framework. Skills therefore affect the choice only through payoffs.<sup>6</sup>

### 3.3 The Worker's Choice

The worker chooses the option with the highest utility, where the utility of option  $j$  is its payoff plus her taste for it,  $U_j = y_j + \tau_j$ . The utility gain from choosing self-employment type  $j \in \{B, A\}$  over wage employment is  $U_j - U_W = \tau_j - c_j$ , where  $c_j \equiv y_W - y_j$  is the opportunity cost of option  $j$ : the payoff she gives up by leaving wage employment. She prefers option

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<sup>6</sup>In reality, tastes may affect how much of each skill a worker accumulates. For example, a worker with a strong preference for autonomy may deliberately choose a career in which she can accumulate more business skills, in anticipation of becoming a business owner. I will discuss this in Section 5.3

$j$  to wage employment when her taste for it exceeds its opportunity cost,  $\tau_j > c_j$ . From equations (1)–(3), the opportunity cost of self-employment type  $j$  is

$$c_j = (1 - r_j^{BX}) BX + (1 - r_j^{SX}) SX, \quad j \in \{B, A\}. \quad (4)$$

Skills thus affect the choice only through these opportunity costs. An increase in skill  $k \in \{BX, SX\}$  changes the opportunity cost of option  $j$  by  $\partial c_j / \partial k = 1 - r_j^k$ . A skill therefore makes option  $j$  less attractive relative to wage employment when option  $j$  pays less for it than wage employment does ( $r_j^k < 1$ ), and more attractive when option  $j$  pays more ( $r_j^k > 1$ ), so its effect on the gain from option  $j$  over wage employment has the sign of  $r_j^k - 1$ .<sup>7</sup>

Figure 1 shows the worker's choice for each possible pair of tastes  $(\tau_B, \tau_A)$ . She chooses wage employment when neither taste exceeds its opportunity cost,  $\tau_B < c_B$  and  $\tau_A < c_A$ . She chooses basic self-employment when  $\tau_B > c_B$  and the gain from basic self-employment exceeds the gain from advanced self-employment,  $\tau_B - c_B > \tau_A - c_A$ , and she chooses advanced self-employment in the opposite case. The three regions meet at the point  $(c_B, c_A)$ . The diagonal boundary between the two types of self-employment is a line with slope one through this point, along which she is indifferent between them,  $\tau_A - c_A = \tau_B - c_B$ .

A worker's location in the figure is given by her tastes. Her skills determine where the boundaries lie, through the opportunity costs  $(c_B, c_A)$ . Because I do not observe her tastes, I treat them as a draw from the density  $g$ . Then, the probability that she chooses option  $j$ ,  $P_j$ , is then the probability that her tastes lie in region corresponding to the option  $j$ . Consider an increase in  $c_B$ . The vertical boundary between wage employment and basic self-employment and the diagonal boundary both shift to the right. If her tastes lie just to the right of the vertical boundary, she now chooses wage employment, and if they lie just below the diagonal boundary, she now chooses advanced self-employment. The probability that she chooses basic self-employment therefore falls, and the probability that she chooses

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<sup>7</sup>If each taste is the sum of a mean and an independent type I extreme value shock, the framework becomes the multinomial logit model of entry that I estimate in Section 5.1. The coefficient on skill  $k$  in the log odds of option  $j$  relative to wage employment is then  $r_j^k - 1$ .

advanced self-employment rises. An increase in  $c_A$  works the same way in the other direction. Hence

$$\frac{\partial P_B}{\partial c_B} < 0, \quad \frac{\partial P_A}{\partial c_B} > 0, \quad \frac{\partial P_B}{\partial c_A} > 0, \quad \frac{\partial P_A}{\partial c_A} < 0. \quad (5)$$

In both cases, the wage employment region grows, so an increase in either opportunity cost lowers the probability that she enters self-employment of either type.

### 3.4 Predictions

The signs of the effects of skills on entry and success depend on the size of the returns. Below, I discuss what the signs would be for different returns, argue which case is most plausible, and state the predictions that follow.

**Business Skills and Self-Employment Entry.** Combining equations (4) and (5), the effect of business skills on the probability of entering each type of self-employment is

$$\frac{\partial P_A}{\partial BX} = \frac{\partial P_A}{\partial c_A} (1 - r_A^{BX}) + \frac{\partial P_A}{\partial c_B} (1 - r_B^{BX}), \quad (6)$$

$$\frac{\partial P_B}{\partial BX} = \frac{\partial P_B}{\partial c_B} (1 - r_B^{BX}) + \frac{\partial P_B}{\partial c_A} (1 - r_A^{BX}). \quad (7)$$

If advanced self-employment pays more for business skills than wage employment does ( $r_A^{BX} > 1$ ), and basic self-employment pays less ( $r_B^{BX} < 1$ ), both terms in equation (6) are positive and both terms in equation (7) are negative. I argue that this is the most plausible case. An owner of an advanced business uses her business skills to coordinate workers, allocate resources, and manage the business as a whole, allowing these skills to generate returns across the entire business. A wage worker, by contrast, applies her business skills within the scope of her assigned responsibilities, limiting the range of activities over which these skills generate returns. In basic self-employment, which involves simpler operations, there is less need for coordination and management, so the returns to business skills are likely to be smaller. Other cases are also possible. If large firms offer managers higher re-

turns to business skills than advanced self-employment does, then  $r_A^{BX} < 1$ , and business skills increase the opportunity cost of both types of self-employment. However, as long as advanced self-employment offers higher returns to business skills than basic self-employment does ( $r_A^{BX} > r_B^{BX}$ ), business skills still shift the boundary between the two types of self-employment toward basic self-employment, making advanced self-employment more likely relative to basic self-employment.

**Prediction 1.** Business skills raise the probability of entering advanced self-employment and lower the probability of entering basic self-employment.

**Specialized Skills and Self-Employment Entry.** An increase in specialized skills changes the opportunity costs of basic and advanced self-employment by  $1 - r_B^{SX}$  and  $1 - r_A^{SX}$ , respectively. If both types of self-employment offer lower returns to specialized skills than wage employment does ( $r_B^{SX} < 1$  and  $r_A^{SX} < 1$ ), specialized skills increase the opportunity cost of both types, reducing the probability of entering either type relative to wage employment. I argue that this is the most plausible case. A business owner's expertise typically applies to only a subset of the tasks required to run the business, whereas a wage worker can devote more of her time to tasks that use her specialized skills. Other cases are possible. When a business relies directly on its owner's expertise, as in a professional practice or skilled trade, self-employment may offer higher returns to specialized skills than wage employment does. The framework therefore predicts a decline in the probability of self-employment relative to wage employment when both returns are lower, but it does not determine the effect on entry into each type separately because an increase in specialized skills may make one type less attractive while shifting workers toward the other.

**Prediction 2.** Specialized skills lower the probability of entering either type of self-employment.

**Skills and Self-Employment Success.** Once the worker is self-employed, the effect of a skill on her payoff is simply the return to that skill,  $r_j^{BX}$  or  $r_j^{SX}$  (equations (2) and (3)). Both skills raise payoffs in both types of self-employment as long as their returns are positive, which is plausible in all cases discussed above. For the reasons given for Prediction 1, business skills are likely to raise payoffs more in advanced than in basic self-employment ( $r_A^{BX} > r_B^{BX}$ ). This comparison also shows the main difference between entry and success. Entry depends on how a return compares with wage employment ( $r_j^k - 1$ ), whereas success depends on the return itself ( $r_j^k$ ). A skill with  $0 < r_j^k < 1$  can therefore lower entry into self-employment and still improve success once the worker is self-employed.

**Prediction 3.** Business skills and specialized skills both raise payoffs in self-employment, and business skills raise payoffs more in advanced than in basic self-employment.

## 4 Data and Measurements

To test these predictions, I use data on workers' skills, entry into self-employment, and outcomes after entry. In this section, I describe the data and how I construct each measure.

### 4.1 Data Sources

The primary data source for this study is the 1979 National Longitudinal Survey of Youth (NLSY79). This survey began in 1979 with a nationally representative sample of 12,686 individuals born between 1957 and 1964 and residing in the United States. The respondents were surveyed annually from 1979 to 1994 and biennially thereafter. I use survey waves from 1979 through 2020.

The NLSY79 is well-suited for my study because it follows respondents over a long horizon. Respondents were ages 14 to 22 at the initial 1979 interview and ages 55 to 63 by the 2020 interview, providing nearly four decades of near-complete work history. This long

coverage is particularly valuable for studying self-employment, which is often entered in the mid or late career. The survey also provides detailed information on individual jobs and businesses, which I discuss next.

For every job held, defined as a unique employer, respondents report start and stop dates (year and month), class of worker, occupation, industry, hours worked, hourly wage, and reason for job separation. The class of worker distinguishes wage employment from self-employment. Self-employed respondents additionally report the incorporation status of the business, which I use to classify different types of self-employment.

Starting in 2010, the NLSY79 introduced a business ownership module that identifies all businesses owned by respondents since age 18. At first encounter with the module, each respondent retrospectively reported every business owned to date, including start and stop dates and company names. Survey staff used this information to link each business to previously reported employers, and these links were verified in the 2012 interview. In each subsequent wave, respondents report any new businesses started since the previous interview, and survey staff continue to link these to contemporaneously reported jobs. This linkage allows me to identify both self-employment and business ownership. I therefore observe business ownership from the start of each respondent’s career through 2020, linked to the job-level record of self-employment. The module also collects detailed information on each business, including the year and month of acquisition, the year the business ended, the amount of financial investment, and typical annual sales or revenue. In addition, beginning in 2010, all respondents—regardless of business ownership—were asked about family business ownership and whether they consider themselves entrepreneurs.

I also use the O\*NET database, version 18.1, administered by the U.S. Department of Labor.<sup>8</sup> O\*NET has collected information on occupations in the United States since 1998, combining surveys of randomly sampled workers with input from occupational experts, and

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<sup>8</sup>I use a single O\*NET version for all years, so each occupation’s skill requirements are held fixed over time. Because O\*NET data start in 1998, matching each NLSY79 survey year with the scores of that time would require combining O\*NET with its predecessor, the Dictionary of Occupational Titles (DOT), which rates different traits on different scales.

version 18.1 covers 873 occupations. Among its domains, I use two: Skills, with 35 elements such as speaking, critical thinking, programming, and repairing, and Knowledge, with 33 elements such as mathematics, design, and transportation. For each occupation, O\*NET rates every element on two dimensions: a level score from 0 to 7, measuring how much of the element the occupation requires, and an importance score from 1 to 5, measuring how central the element is to the occupation. I multiply the two scores and refer to the product as the element score.<sup>9</sup> The element scores for the 68 elements form the basis for the three skill measures—business skills, specialized skills, and skill balance—which are the main independent variables in the analysis. O\*NET also assigns each occupation to one of five job zones, which classify occupations by their typical requirements for education, related experience, and on-the-job training. I use job zones in a robustness analysis to assess whether business skills capture workers’ accumulated experience in occupations requiring greater preparation, rather than a distinct set of business skills.

I merge the NLSY79 and O\*NET datasets using a crosswalk between their occupational classification systems. The NLSY79 uses the 1970 Census occupation codes through 2000 and the 2000 Census codes thereafter, while O\*NET uses the 2010 Standard Occupational Classification (SOC). To link both NLSY79 coding schemes to the 2010 SOC, I use the crosswalks developed by Lise and Postel-Vinay (2020).

## 4.2 Sample Construction

I construct separate samples for the entry analysis and the success analysis because the two require different data structures. The entry decision is modeled as year-by-year transitions constructed from job start and stop dates. Success after entry is measured using business revenue and earnings, which are reported at the interview date. Revenue is reported in two separate modules—the business ownership module and, from 2010 onward, the employer

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<sup>9</sup>The element score takes into account both how much of an element the occupation requires and how central the element is to the work. In practice, the two scores are highly correlated: across the occupations held by workers in my sample, the correlation between level and importance is 0.96, and each has a correlation of 0.94 or more with their product.

supplement—which raises additional sample selection issues that I discuss below. All samples share the same set of individual-level selection criteria, described in the sample for the entry analysis.

**Sample for Entry Analysis** The entry analysis uses a person-year panel to estimate the competing-risks multinomial logit (MNL) model described in Section 5.1. The model captures workers’ year-by-year decisions to remain in non-self-employment or enter different types of self-employment. To construct this panel, I first identify each respondent’s career start date, defined as the beginning of the first nonenrollment spell lasting at least twelve months, and then construct career histories using job start and stop dates. Because the entry analysis requires observing both employment transitions and subsequent self-employment entry, I impose the following individual-level sample restrictions.

Starting from the 12,686 NLSY79 respondents, I first exclude 392 individuals who leave the survey before their career start date or before holding at least one job after the start date. Second, I remove 936 individuals with incomplete job start or stop dates, which prevents the construction of a person-year panel. Third, I drop 4,037 individuals who exit the survey before 2010, when the business ownership module was introduced. Fourth, I exclude 25 individuals who are already self-employed within the first twelve months of their careers and remain self-employed throughout the sample period. Finally, I remove 180 individuals whose career histories exceed 45 years. These individuals entered the labor market at unusually young ages, and information on their employment prior to the 1979 survey is limited. These restrictions leave a sample of 7,116 individuals.

I then expand the career histories of these individuals into a person-year panel, with one observation for each twelve-month period from career start until self-employment entry or the final interview. The analysis focuses on periods before self-employment entry, when individuals are at risk of transitioning from non-self-employment—which includes wage employment, unemployment, and being out of the labor force—into self-employment. Individuals who

return to non-self-employment after a self-employment spell reenter the sample and contribute additional observations. The final panel contains 251,214 person-year observations from 7,116 individuals.

**Sample for Success Analysis** The success analysis examines the relationship between skills and business performance using two samples: a business revenue sample and an earnings sample. Both are constructed from self-employment jobs observed in the entry sample, with the self-employment job-year as the unit of observation.

The business revenue sample is restricted to self-employment job-years with valid revenue information. Revenue comes from two sources: the business ownership module, which reports typical annual sales or revenue for each business, and the employer supplement, which reports revenue repeatedly from 2010 onward. I prioritize the employer supplement, which provides contemporaneous revenue information, and use the business ownership module as a fallback. Revenue is available only for jobs that respondents report as a business, so self-employment jobs identified only by class of worker are excluded. The resulting sample contains 2,430 basic and 2,465 advanced self-employment job-year observations.

The earnings sample includes all self-employment job-years in the entry sample with reported weekly earnings, including jobs identified only by class of worker. It therefore covers more spells than the revenue sample and contains 10,383 observations from 4,118 basic self-employment jobs and 5,938 observations from 1,554 advanced self-employment jobs. For a direct comparison between revenue and earnings, I also restrict the earnings analysis to the revenue sample, allowing both outcomes to be examined over the same set of job-years. The revenue sample represents approximately 23 percent of the earnings sample for basic self-employment and 42 percent for advanced self-employment. This difference reflects the lower likelihood that self-employed workers who engage in small, basic operations report their work as a business.

### 4.3 Classifying Basic and Advanced Self-Employment

I define a job as self-employment if either its class of worker is recorded as self-employed in the employer history or it can be linked to a business the respondent reports owning in the business ownership module. Using both sources helps capture two groups of owners who might otherwise be missed: owners of incorporated businesses who report themselves as employees of their own firms, and owners of small, home-based businesses who do not report their work as a business (Light and Munk, 2018). For the few businesses reported in the business ownership module that cannot be linked to a job in the employer history, I exclude them because the analysis samples are constructed from the job panel.

I distinguish between basic and advanced self-employment using two measures of investment or business commitment at entry: incorporation status and start-up capital. Incorporation status captures the legal form of the business, while start-up capital measures the financial resources committed to it. Incorporation may reflect tax or liability considerations, and businesses with the same legal form at entry can differ substantially in their initial investment (Figure 2). I therefore use both measures to account for the heterogeneity in self-employment.

A self-employment spell is classified as basic if the business is unincorporated and its start-up capital is less than 5,000 USD in 2010 dollars. A spell is classified as advanced if the business is incorporated or its start-up capital is at least 5,000 USD. The start-up capital cutoff is the median among spells with positive initial start-up capital. Of the 4,566 self-employment spells in the entry sample, 76 percent are classified as basic and 24 percent as advanced.<sup>10</sup>

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<sup>10</sup>Another common approach to classify self-employment is to consider firm size, or the number of employees. I do not use this because firm size was not collected from 1981 to 1985 in the NLSY79, which may lead to inconsistent measurement across career stages.

## 4.4 Outcome Variables

I examine two outcomes: entry into self-employment and business success conditional on entry. I measure business success along two dimensions: business revenue and earnings.

### **Self-Employment Entry: Transition into Basic or Advanced Self-Employment**

The dependent variable  $Y_{it}$  records the individual's transition over the next 12 months among three possible outcomes: remaining in non-self-employment ( $Y_{it} = 0$ ), entering basic self-employment ( $Y_{it} = 1$ ), or entering advanced self-employment ( $Y_{it} = 2$ ). Transitions between jobs within self-employment are not counted as new entries. Figure 3 illustrates an example of how I construct this outcome for a worker across the years in which she is at risk of entering self-employment.

**Self-Employment Success: Business Revenue and Earnings** I use two outcomes that capture different aspects of self-employment success after entry: business revenue and earnings. Business revenue measures the sales generated by the business, while earnings measure the income the owner takes home. When respondents report earnings on a weekly basis, I use the reported amount directly. For other payment frequencies, I calculate weekly earnings by multiplying reported hourly wages by hours worked per week. I use weekly rather than hourly earnings because self-employed workers can adjust their hours more flexibly, so differences in earnings may reflect both hourly returns and hours worked. Both revenue and earnings are adjusted to 2010 USD using the annual Consumer Price Index for the four United States regions (Northeast, Midwest, South, and West).

## 4.5 Independent Variables

In this section, I first discuss the three O\*NET based skill measures—business skills, specialized skills, and balanced skills—which are the main independent variables. I then explain how I construct occupation count, a conventional measure of skill balance, and baseline

regressors included in the empirical specifications.

#### 4.5.1 Business Skills

To construct the business skills measure ( $BX$ ), I combine information on the business-skill intensity of occupations with workers' occupational histories. The construction is done in three steps. First, I identify the O\*NET elements that capture business skills. Second, I use these elements to measure the business-skill intensity of each occupation. Third, I combine these occupational intensities with workers' accumulated experience in each occupation to construct  $BX$ .

I begin by identifying O\*NET elements that capture three components of establishing and operating a business: core business knowledge, managerial skills, and interpersonal skills. Core business knowledge includes economics and accounting, sales and marketing, production and processing, customer and personal service, and service orientation. Managerial skills consist of administration and management, personnel and human resources, and the management of financial, material, and personnel resources. Interpersonal skills consist of social perceptiveness, coordination, persuasion, negotiation, and instructing. In making this selection, I use the O\*NET elements that rank highest for chief executives (CEOs) as a reference, because the CEO occupation encompasses key activities of business ownership, including making financial and strategic decisions, managing staff and resources, and interacting with customers or clients. This process yields 15 of the 68 elements, which I refer to as business elements. Table A1 lists all 68 elements and identifies the 15 selected to construct the business skill measure.

Next, I use these 15 business elements to assign each occupation a business-skill intensity. Specifically, I calculate the average element score of the 15 business elements for each occupation and normalize this average by the highest value across all occupations. This normalization sets the most business-intensive occupation to 1 and expresses the intensity of every other occupation relative to that benchmark. The CEO occupation has the highest

average scores, 4.76, and therefore serves as the benchmark.

Finally, I combine these occupational intensities with workers' occupational histories to construct the business skills measure,  $BX$ . For worker  $i$  at time  $t$ , I define

$$BX_{it} = \sum_{o \in O_i} \omega_o^{BX} \cdot OccExp_{iot}, \quad (8)$$

where  $O_i$  is the set of occupations the worker  $i$  has ever held, and  $OccExp_{iot}$  is her cumulative experience in occupation  $o$  up to time  $t$ , measured in years. The weight  $\omega_o^{BX}$  is the normalized business-skill intensity of occupation  $o$ . Thus, experience in occupations with higher business-skill intensity contributes more to  $BX$ , while experience in occupations with lower business-skill intensity contributes less. Because the CEO occupation defines the benchmark, one year of experience as a CEO contributes one unit to  $BX$ .

For example, ten years of uninterrupted employment as a CEO yield a  $BX$  of 10. Ten years as a purchasing manager, whose average business-element score is 4.09, give a  $BX$  of  $(4.09/4.76) \times 10 = 8.6$ . By contrast, ten years as an agricultural product grader or sorter, one of the least business-intensive occupations, with an average score of 1.10, give a  $BX$  of  $(1.10/4.76) \times 10 = 2.3$ . These examples illustrate how  $BX$  assigns greater weight to experience acquired in occupations with higher business-skill intensity.

A potential concern is that  $BX$  captures general skill requirements rather than business-specific skills, since occupations that require and use more business skills are likely to also require more education, training, and prior experience. To assess this concern, I re-estimate the entry models controlling for O\*NET job zones, a five-level classification of the preparation required for an occupation, ranging from Zone 1 (little or no preparation) to Zone 5 (extensive preparation). Table A4 reports the estimated marginal effects with and without these controls. The estimated effects of business skills remain similar in magnitude across the two specifications, suggesting that  $BX$  captures skills specific to establishing and operating a business rather than simply reflecting the overall skill level.

### 4.5.2 Specialized Skills

Specialized skills measure the extent to which a worker’s experience matches the specific expertise required by her target business. For example, a worker who starts a law firm needs expertise in law and government, whereas a worker who starts an electrical contracting business needs expertise in building and construction and mechanical work. Because the expertise required varies across businesses, the measure is tailored to each worker’s target occupation.

I identify each worker’s area of specialization using her *target occupation*. For workers who enter self-employment, the target occupation is the occupation held at the start of the self-employment spell. For workers who never enter self-employment, the target occupation is the occupation held eighteen years into the career, corresponding to the average point of entry into self-employment in the sample.<sup>11</sup>

For each target occupation, I rank all 68 O\*NET elements by their element scores, defined as the product of the importance and level scores. I select the top one to three elements, using natural breaks in the ranking to identify elements that stand out from the rest. I then exclude elements that rank near the top across most occupations, since broadly required skills do not distinguish a worker’s area of specialization. For example, the top three elements for lawyers are law and government, English language, and speaking. Because English language and speaking rank near the top in many occupations, I keep only law and government as the specialization element. For electricians, the top two elements are building and construction and mechanical, and both are kept.<sup>12</sup>

I construct specialized skills using the same general approach as the business-skills mea-

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<sup>11</sup>The results are robust to instead defining the target occupation of these workers as the occupation in which they have spent the most time.

<sup>12</sup>This procedure is similar in spirit to TF-IDF (term frequency–inverse document frequency) weighting in natural language processing, where element scores capture the strength of an element within an occupation, while excluding elements that rank near the top across most occupations removes elements that are not distinctive to a particular occupation.

sure, but with weights tailored to each worker’s target occupation as follows:

$$SX_{it} = \sum_{o \in O_i} \omega_{io}^{SX} \cdot OccExp_{iot}, \quad (9)$$

where the weight  $\omega_{io}^{SX}$  is the average element score, in occupation  $o$ , of the elements selected as worker  $i$ ’s specialization elements, normalized by the maximum value across occupations. Thus, an occupation receives a weight of 1 if it uses worker  $i$ ’s specialization elements most intensively. Because the weights depend on the target occupation, two workers with identical work histories can have different values of  $SX$  if their target occupations differ. The measure therefore captures experience in occupations that develop the specific expertise required by a worker’s target business, rather than work experience in general.

#### 4.5.3 Skill Balance

I construct a direct measure of skill balance using O\*NET to reexamine the JOAT hypothesis. The measure, which I call the *balanced skill count* ( $BSC$ ), captures how evenly a worker’s accumulated skill portfolio is distributed across O\*NET elements.

To construct the skill portfolio for worker  $i$  at time  $t$ , I calculate the accumulated amount of each O\*NET element by summing its element scores across all occupations the worker has held, weighted by experience in each occupation. I then retain the 30 elements with the largest accumulated amounts out of the 68 O\*NET elements. Let  $p_{its}$  denote the share of element  $s$  in the worker’s total accumulated amount across these 30 elements.<sup>13</sup> I summarize the distribution using the exponentiated Shannon index:

$$BSC_{it} = \exp \left( - \sum_s p_{its} \log p_{its} \right). \quad (10)$$

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<sup>13</sup>I exclude the remaining 38 elements because workers typically accumulate little or no experience in some of them. Including all 68 would mechanically make every worker’s portfolio appear unbalanced, even when her accumulated skills are evenly distributed across the elements relevant to her work experience. On average, the top 30 elements account for about 80 percent of a worker’s accumulated total skills across all 68 elements.

The *BSC* can be interpreted as the effective number of elements in a worker’s skill portfolio, which is the number of elements that would produce the same degree of balance if all received equal shares. It ranges from 1, when all accumulated skill is concentrated in a single element, to 30, when it is distributed equally across all 30 elements. A higher *BSC* therefore indicates a more balanced skill portfolio, while a lower *BSC* indicates greater concentration in a few elements.

Consider two workers, each with 15 years of experience divided equally across three employers. The first has worked as a bookkeeping clerk, an accountant, and a financial analyst. Because these occupations draw heavily on overlapping elements, such as mathematics and economics and accounting, her accumulated skill portfolio is concentrated in a relatively small set of elements. The second has worked as an accountant, a registered nurse, and a construction supervisor. These occupations draw on more distinct elements, spanning economics and accounting, medicine and dentistry, and building and construction. Her accumulated skill portfolio is therefore more evenly distributed across elements, resulting in a higher *BSC*, even though a count of employers assigns the two workers the same value.

#### 4.5.4 Occupation Count

To reconcile my findings with the existing JOAT literature, I also use a conventional proxy for skill balance, which is the cumulative number of distinct occupations an individual has held. I classify occupations using the 23 major groups of the 2010 Standard Occupational Classification (SOC), defined by the first two digits of the SOC code linked to each NLSY79 occupation. The count increases when a worker enters a major occupational group she has not previously worked in, and returning to a previously held group does not increase the count. Unlike business skills, specialized skills, and balanced skill count, this measure does not use O\*NET.

I also construct a cumulative measure of involuntary job separations to assess whether the occupation count captures job churn rather than true skill balance. For each job that

ends, the NLSY79 records the reason for separation. I classify a separation as involuntary if the worker was laid off, the plant closed, a temporary or seasonal job ended, the worker was discharged or fired, or a program ended. I then sum these separations from the start of the worker’s career through time  $t$ .

#### 4.5.5 Baseline Regressors

In my empirical analysis, I include a rich set of control variables. The choice of controls follows a large literature on the determinants of business ownership, including risk tolerance (Ahn, 2010; Hsieh, Parker and van Praag, 2017), access to credit and household wealth (Hurst and Lusardi, 2004; Fairlie and Krashinsky, 2012), and family background (Hout and Rosen, 2000; Hvide and Oyer, 2025).

I first control for past work experience, measured in two ways. Potential experience is the time elapsed since the career start date. Actual experience is the time actually worked, adjusted for the weekly hours worked on each job. Both enter with their squares. I also control for demographics (gender, race/ethnicity, cohabitation and marital status, and whether the youngest child in the household is age six or younger<sup>14</sup>).

I then control for risk tolerance. Risk tolerance is a four-category measure built from the income-gamble question in the 1993 and 2002–2014 NLSY79 waves, with missing values linearly interpolated. I also consider personality traits including risk tolerance, sociability at age 6 (four categories from extremely shy to extremely outgoing), Rotter’s locus of control score (from 4 to 16), and three self-rated items on a 1-7 scale capturing how extraverted/enthusiastic, open to new experiences/complex, and conventional/uncreative the respondent describes herself.

Next, I control for family resources using total net family wealth, net family income, whether the respondent’s father had at least some college education, and the father’s one-digit occupational category in 1979. Because net family wealth is available only from 1985

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<sup>14</sup>Having a young child can motivate entry into self-employment as a means to balance work and family, especially for women. See Lim (2019); Lim and Michelsmore (2018).

onward, I impute wealth for earlier years using predicted values from a linear model that includes age, age squared, gender, race/ethnicity, marital status, and educational attainment.

I also control for education (less than high school, high school graduate, some college, and college or above) and AFQT percentile as a proxy for cognitive ability. Finally, I control for whether any family member owned a business and for the number of past self-employment spells. I also use whether the respondent considers herself an entrepreneur as a proxy for her taste for self-employment. Because this measure is available only from 2010 onward and may reflect prior self-employment experience, I exclude it from the baseline specification and examine its inclusion in a sensitivity analysis in Section 5.3.

## 4.6 Data Patterns

Table 1 reports the distribution of individuals in the entry sample by the number of self-employment entries they make (Panel A) and the types of self-employment they enter (Panel B). Most individuals never enter self-employment (58.5 percent), and among those who do, 88 percent enter only once or twice. Basic self-employment is the more common destination, with 27.9 percent of individuals entering only basic self-employment, compared with 8.4 percent entering only advanced self-employment. A further 5.2 percent enter both types at different points in their careers, representing about one in eight individuals who ever enter self-employment.

Table 2 reports means and standard deviations by employment type. The sample consists of the job spells in the entry sample, and each spell is measured at its start. Panel A shows that basic and advanced self-employment differ sharply. An advanced business has about 212,000 USD in revenue, roughly four times the 55,000 USD of a basic business, and its owner earns about twice as much per week. Advanced spells also last longer, averaging 7.1 years compared with 3.5 years for basic spells. Because many advanced self-employment spells are still ongoing at the last interview these durations understate how long spells last. Advanced spells are also more concentrated in management, business, science, and arts occupations.

These gaps justify treating basic and advanced self-employment as qualitatively different types rather than collapsing them into a single category.

Work experiences, O\*NET-based skill measures, and indicators of job churn are shown in Panel B. Entry into self-employment typically occurs in mid-career. Advanced entrants have accumulated more actual work experience than basic entrants, at 15.24 versus 10.72 years. This pattern holds even when I restrict the focus to first self-employment spells. Although not reported, I find that among those who become self-employed, the first basic self-employment starts at age 32 on average, about 13 years into the career, and the first advanced business at age 37, about 16 years in. Advanced entrants have also accumulated more experience in both business and specialized skills than basic entrants. In contrast, skill balance is nearly identical across the three groups. Advanced entrants also exhibit less job churn than basic entrants. Per year of actual experience, advanced entrants have worked in fewer major occupation groups and experienced fewer involuntary job losses. These patterns suggest that occupation counts may capture not only the accumulation of different work experience but also differences in career instability, motivating the distinction between skill balance and job mobility.

In Panel C, I summarize individual characteristics. Basic and advanced self-employment entrants differ along several dimensions. Advanced entrants are more likely to be male, married, and college-educated, score higher on the AFQT, come from wealthier families with a family business background, and identify as an entrepreneur. Differences in personality traits, such as sociability at age 6, Rotter locus of control, and the three self-rated traits are more modest.

Table 3 presents partial correlations among the three O\*NET-based skill measures, conventional occupation count, and number of involuntary job losses, calculated at the start of a job spell and controlling for actual work experience, separately by employment type. The correlations are particularly informative about how the O\*NET-based skill measures relate to one another and how conventional occupation counts compare with these measures.

First, business skills and specialized skills are positively correlated across all employment types, with correlations ranging from 0.398 to 0.563, but the relationship is far from perfect. The two measures therefore capture some overlapping but distinct dimensions of human capital. Second, skill balance is weakly correlated with either skill measure. Its correlation with business skills is small and positive, reaching at most 0.153, while its correlation with specialized skills is close to zero. Third, the number of major occupation groups held is positively correlated with skill balance, which is consistent with the conventional interpretation of occupational count as a proxy for balanced skills. However, the count is also positively correlated with involuntary job losses and negatively correlated with both business and specialized skills. These relationships suggest that occupation counts capture more than skill balance alone, reflecting differences in job mobility. Therefore, occupation counts may serve as an imperfect proxy for skill balance, motivating my use of the O\*NET-based measures to distinguish skill balance from job mobility.

## 5 Empirical Strategy

In this section, I describe the empirical strategies used to test the predictions of my conceptual framework for self-employment entry and post-entry success (business revenue and earnings).

Before turning to each empirical strategy, I explain a common scale for the skill measures and a consistent approach to reporting their estimated effects. I standardize business skills ( $BX$ ), specialized skills ( $SX$ ), and skill balance ( $BSC$ ) to have a mean of zero and a standard deviation of two. Thus, a half-standard-deviation increase corresponds to a one-unit increase in each standardized measure; for  $BX$  and  $SX$ , this is roughly equivalent to two additional years of the corresponding skill. Throughout, I report average marginal effects (AMEs) associated with a half-standard-deviation increase in each skill measure. AMEs facilitate interpretation in the nonlinear entry models and accommodate the quadratic specification,

under which the effect of each skill varies with its level rather than being captured by a single coefficient.

## 5.1 Entry into Self-Employment

I model entry into self-employment as a discrete-time competing-risks problem. In each year, a worker who is not currently self-employed faces three options. She can move into basic self-employment, move into advanced self-employment, or stay out of self-employment. I estimate this as a multinomial logit. The latent value of option  $j$  for worker  $i$  in year  $t$  is

$$V_{ijt} = \alpha_{0j} + \alpha_{1j}BX_{it} + \alpha_{2j}BX_{it}^2 + \alpha_{3j}SX_{it} + \alpha_{4j}SX_{it}^2 + \alpha_{5j}BSC_{it} + \alpha_{6j}BSC_{it}^2 + X_{it}\alpha_{Xj} + \varepsilon_{ijt}, \quad (11)$$

where  $BX_{it}$  is business skills,  $SX_{it}$  is specialized skills, and  $BSC_{it}$  is the balanced skill count, and  $X_{it}$  is the set of baseline regressors from Section 4.5.5. The worker chooses the option with the highest latent value. The coefficients depend on  $j$  because each skill measure and each covariate can affect entry into basic and advanced self-employment differently. I also allow for a nonlinear relationship between each skill measure and entry by including a quadratic term. Standard errors are clustered at the individual level to account for correlation over time for a given individual.

## 5.2 Success in Self-Employment

I estimate success models separately for basic and advanced self-employment because the conceptual framework predicts that the two types reward skills differently. Because these models include only workers who enter self-employment, they characterize the relationship between skills and self-employment outcomes conditional on entry. In the conceptual framework, these relationships correspond to the skill-specific returns,  $r_B^{BX}$ ,  $r_B^{SX}$ ,  $r_A^{BX}$ , and  $r_A^{SX}$ .

I first examine the relationship between skills and annual business revenue using a Tobit

model to account for top-coding in the NLSY79. The dependent variable is log annual business revenue. As in the entry models, the regressors include  $BX$ ,  $SX$ , and  $BSC$ , each entered with a quadratic term. In addition to the baseline regressors, I also control for tenure in the current self-employment job and its square, and region and industry fixed effects. Standard errors are clustered at the individual level.

To compare business performance with the earnings owners receive, I next estimate an OLS model with log weekly earnings as the dependent variable, using the same sample of self-employment spells with observed revenue. Holding the sample the same allows me to compare the two outcomes directly. Revenue captures the performance of the business itself, whereas earnings capture the income the owner takes home. The two outcomes thus reflect different aspects of self-employment success and need not exhibit the same relationship with skills.

Finally, I estimate the earnings specification on the broader sample of self-employment spells with observed earnings, including those without reported revenue. This specification allows me to assess whether the earnings results using the revenue sample extend beyond the subset of self-employment spells for which revenue is observed.<sup>15</sup>

### 5.3 Selection on Observables

A concern for interpreting the estimated relationships between skills and self-employment outcomes is that accumulated skills may reflect not only workers' human capital but also their preferences and other unobservable characteristics. For example, a worker who values autonomy and plans to start a business may select occupations that develop business skills. In this case, business skills may be positively associated with self-employment entry partly because they reflect the worker's preexisting preferences, rather than solely because they make self-employment more attractive.

This concern is particularly relevant for entry in the conceptual framework. The taste

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<sup>15</sup>Another approach to examine how skills affect earnings would be compare the earnings between different employment types within the same worker using individual fixed effects. I discuss this in Appendix C.

parameters  $\tau_B$  and  $\tau_A$  capture factors other than skills that affect the attractiveness of basic and advanced self-employment, respectively, including observables such as wealth, risk tolerance, personality traits, and unobservables such as preferences for autonomy and business aspirations. Although the framework treats skills and tastes as given, tastes may also shape skill accumulation in practice. Consequently, the estimated association between skills and entry may partly reflect these underlying characteristics. A similar concern applies to the success models. Among workers who enter self-employment, unobserved characteristics may be associated with both accumulated skills and subsequent business revenue and earnings, even though tastes do not directly enter the payoff functions in the conceptual framework.

My empirical strategy therefore relies on a selection-on-observables assumption: conditional on the included controls, the remaining determinants of self-employment entry and success are unrelated to the skill measures. The controls described in Section 4.5.5 capture observable characteristics that may be related to both workers' preferences and their skill accumulation. Although the assumption cannot be tested directly, I assess the sensitivity of the estimated skill coefficients to the sequential addition of control variables, focusing on the entry models.

I also consider two alternative approaches, discussed in more detail in Appendix B. The first is a conditional logit model with individual fixed effects, which absorbs time-invariant unobserved heterogeneity. However, it cannot account for unobserved characteristics that change over a worker's career, and its estimates are imprecise in my application because skills accumulate gradually, leaving limited within-person variation. The second approach uses instrumental variables, exploiting variation in the industrial composition of a worker's county at labor market entry to instrument for accumulated business and specialized skills. However, the instruments are weak, and the two instruments are too highly correlated to separately identify the effects of the two skill measures. I therefore leave both approaches for future work.

## 6 Findings

The findings reveal how business skills, specialized skills, and skill balance relate to entry into self-employment and subsequent success, and how these relationships compare with the predictions of the conceptual framework in Section 3.4. I first present results for entry into basic and advanced self-employment, then turn to success after entry, measured by revenue and earnings.

### 6.1 How Skills Predict Entry into Self-Employment

Table 4 reports estimated marginal effects of business and specialized skills on entry into basic self-employment (Panel A) and advanced self-employment (Panel B). Columns 1 and 2 include each skill measure separately, while column 3 includes both jointly. The estimates show opposite patterns across the two skill measures. In the joint specification, a half standard deviation increase in business skills is associated with a 34.5 percent higher probability of entering advanced self-employment and a 13.5 percent lower probability of entering basic self-employment. By contrast, a half standard deviation increase in specialized skills is associated with a 4.6 percent lower probability of entering advanced self-employment and a 13.7 percent lower probability of entering basic self-employment.

These patterns support Predictions 1 and 2. The positive association between business skills and entry into advanced self-employment, together with the negative association with entry into basic self-employment, is consistent with the framework’s prediction that business skills are rewarded more in advanced self-employment than in wage employment, but less in basic self-employment ( $r_A^{BX} > 1 > r_B^{BX}$ ). This pattern is consistent with the framework’s prediction that business skills are rewarded more in advanced self-employment, where they are directly useful for running a business, but less in basic self-employment, where workers with strong business skills may have better outside options.

The negative associations between specialized skills and entry into both types of self-

employment are also consistent with the framework’s prediction that specialized skills are rewarded less in both types of self-employment than in wage employment ( $r_B^{SX} < 1$  and  $r_A^{SX} < 1$ ). Greater specialization therefore raises the opportunity cost of leaving wage employment, discouraging entry into either type of self-employment.

Next, I revisit Lazear’s JOAT hypothesis, which predicts that workers with more balanced skills are more likely to become business owners, using my direct measure of balanced skills. Column 4 of Table 4 includes *BSC* on its own. Skill balance is positively associated with entry into advanced self-employment, with a modes estimated effect of 7.9 percent per half standard deviaion, but is unrelated to entry into basic self-employment. Column 5 adds skill balance to the specification that already includes business skills and specialized skills. In this specification, skill balance is no longer significantly associated with entry into either type of self-employment. Thus, although skill balance predicts entry into advanced self-employment when considered along, it provides no additional predictive power beyond the types of skills workers have accumulated.

To reconcile these findings with the existing JOAT literature, I first replicate the conventional empirical approach by pooling basic and advanced self-employment into a single category and estimating a logit model. Column 6 of Panel C shows that the occupation count positively predicts entry into self-employment, consistent with the existing literature. However, this pooled result masks important differences across the two types of self-employment. When I separate the types of self-employment as shown in Panels A and B, the estimated association is substantially stronger for basic self-employment: a half standard deviation increase in the occupation count is associated with a 14.8 percent higher probability of entry, compared with only 1.2 percent for advanced self-employment. Moreover, when I include the direct measures of business skills, specialized skills, and balanced skills in column 7, the occupation count is no longer significantly associated with entry into advanced self-employment, although it remains predictive of entry into basic self-employment. These results suggest that the conventional positive association between occupation count and self-employment is

driven primarily by entry into basic self-employment.

The remaining association between occupation count and entry into basic self-employment may reflect job churning. To examine this possibility, I add the cumulative number of involuntary job losses in column 8. Involuntary job loss are more strongly associated with entry into basic self-employment than advanced self-employment. A half standard deviation increase is associated with an 11.3 percent higher probability of entry into basic. Including this measure also reduces the estimated association between occupation count and entry by about one third. This attenuation suggests that occupation count partly captures employment instability, which may lead workers turn to unincorporated, low-capital self-employment.

Taken together, the entry results so far refine the conventional JOAT hypothesis in three ways. First, entry depends on the types of skills workers have accumulated—business skills and specialized skills—rather than skill balance. Second, pooling basic and advanced self-employment obscures important differences in how skills predict entry. Business skills have opposite associations with entry into the two types, while the conventional positive association between occupation count and self-employment is driven primarily by basic self-employment. Third, occupation count partly reflects job churning rather than skill balance. Its association with entry into basic self-employment falls by about one third when involuntary job losses are controlled for. Together, these findings suggest that the conventional empirical relationship between occupational count and any type of self-employment does not necessarily reflect the role of balanced skills in preparing workers for business ownership.<sup>16</sup>

I end this subsection by examining the sensitivity of the entry estimates to controlling for additional determinants of self-employment, as discussed in Section 5.3. In Table 5, I start with a specification where the three O\*NET skill measures, potential experience, and actual

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<sup>16</sup>My findings also help reconcile evidence in the JOAT literature that has been difficult to explain. Lechmann and Schnabel (2014) find that occupation counts predict entry into solo self-employment but not self-employment with employees, while Åstebro and Thompson (2011) find that workers with more occupations tend to earn less in self-employment. Both findings are not consistent with the JOAT hypothesis, but are consistent with my results if solo self-employment more closely resembles basic self-employment, where occupation counts partly reflect job churning, and earnings are substantially lower than in advanced self-employment (Table 2).

experience are taken into account. Then, I sequentially add demographic characteristics, personality traits, resources, education and cognitive ability, family business background, and past self-employment spells. The estimates of business skills retain their predicted signs and remain statistically significant across all specifications, although the magnitudes change. For advanced self-employment, the estimated marginal effect falls from 41.9 percent in column 1 to 33.6 percent in column 7 which is the baseline specification (identical to column 5 of Table 4). The largest reduction occurred following the inclusion of resources (total net family income and total net family wealth) and personality traits. Adding family business ownership and past self-employment spells show little additional effect, while controlling for entrepreneurial identity reduces the estimate further to 22.8 percent (column 8). Because entrepreneurial identity is measured only from 2010, many respondents had already entered self-employment by the time they were asked whether they considered themselves entrepreneurs. Their responses may therefore reflect their self-employment experience, rather than only their pre-existing aspirations to start a business. Column 8 should therefore be interpreted with caution, as controlling for entrepreneurial identity may capture differences that emerged after workers entered self-employment. For basic self-employment, the negative estimated marginal effects of business skills become stronger as controls are added, reaching  $-14.2$  percent in the baseline. The estimates for specialized skills remain largely unchanged across specifications except for accounting demographics in advanced self-employment entry. Overall, the results suggest that differences in demographics, personality traits, resources, and entrepreneurial identity account for part, but not all, of the association between business skills and entry into different types of self-employment.

## 6.2 How Skills Predict Success in Self-Employment

Business skills and specialized skills are associated with higher payoffs in self-employment, but these relationships differ by type of self-employment. Business skills are more strongly associated with revenue and earnings in advanced than in basic self-employment, while special-

ized skills are more strongly associated with revenue and earnings in basic self-employment. Skill balance, by contrast, shows little relationship with success.

Table 6 reports the estimated marginal effects. Column 1 reports Tobit estimates for log annual revenue, column 2 reports OLS estimates for log weekly earnings on the same revenue sample, and column 3 reports OLS estimates for log weekly earnings on the broader sample of all self-employment job-years with reported earnings.

Business skills are more strongly associated with success in advanced than in basic self-employment. In advanced self-employment, a half standard deviation increase in business skills is associated with 39.4 percent higher revenue and 14.2 to 16.3 percent higher earnings. The revenue estimate is significant at the 1 percent level, and the earnings estimate in the broader sample is significant at the 5 percent level. The earnings estimate in the revenue sample is similar in size but less precise ( $p = 0.11$ ). In basic self-employment, the same increase is associated with 21.7 percent higher revenue, significant at the 10 percent level, but with small and statistically insignificant differences in earnings. In both types of self-employment, the estimated association is larger for revenue than for earnings. One possible explanation is that owners with stronger business skills expand their businesses by reinvesting profits rather than increasing the income they take home. Earnings may therefore understate the contribution of business skills to business performance.

Specialized skills are positively associated with success in basic self-employment, despite discouraging entry into it. A half standard deviation increase in specialized skills is associated with 17.3 percent higher revenue and 14.9 to 19.2 percent higher earnings in basic self-employment, with all estimates significant at the 1 percent level. In advanced self-employment, the estimates are smaller and less precise, at 10.7 percent for revenue and 8 percent for earnings. The earnings estimate in the broader sample is marginally more precise but remains statistically insignificant at conventional levels ( $p = 0.11$ ). Unlike business skills, specialized skills have similar estimated associations with revenue and earnings, suggesting that their contribution to business performance may translate more directly into owners'

earnings.

Skill balance shows little relationship with success. Its estimated associations with revenue and earnings in the revenue sample are statistically insignificant in both types of self-employment. In the broader earnings sample, skill balance is associated with 3.1 percent higher earnings in basic self-employment per half standard deviation increase, but the estimate is small and this pattern does not appear in the revenue results.

These findings are broadly consistent with Prediction 3. Business skills are more strongly associated with success in advanced than in basic self-employment, while specialized skills are positively associated with success in both types, with stronger and more precisely estimated relationships in basic self-employment. Although the estimates for specialized skills in advanced self-employment are not statistically significant, their positive signs are consistent with the prediction that both types of skills raise payoffs in self-employment. The positive association between specialized skills and success in basic self-employment, despite their negative association with entry, also illustrates a key distinction in the framework: entry depends on a skill's return relative to its return in wage employment, whereas success depends on its return within self-employment. A skill can therefore discourage entry while raising payoffs for those who enter.

## 7 Conclusion

I study what kind of work experience prepares someone to start a business. I extend the JOAT literature in three ways: I ask which type of skills a worker has accumulated rather than how balanced they are, I separate incorporated, higher capital businesses from unincorporated, low-capital work, and I measure skill balance directly rather than by counting occupations. I develop a conceptual framework where business skills and specialized skills are rewarded differently across employment types, generating predictions for both entry and success after entry. Using NLSY79 work histories linked to O\*NET, I construct time-in-

occupation-weighted measures of business skills, specialized skills, and skill balance, and test predictions derived from the framework using self-employment entry, revenue, and earnings outcomes.

The findings are consistent with the conceptual framework’s distinction between the skills that make self-employment attractive relative to wage employment and those that contribute to success after entry. Business skills are associated with entry into advanced self-employment rather than basic, where their relationship with subsequent revenue and earnings is also strongest. Specialized skills, by contrast, are associated with lower entry into both types of self-employment but with greater success among those in basic self-employment. This difference reflects the role of opportunity costs in occupational choice—specialized workers may have more to gain from remaining in wage employment even when their skills are productive in self-employment. The findings thus highlight why the skills associated with entering self-employment need not be the same as those associated with success once self-employed.

These findings refine the JOAT hypothesis in three ways. First, directly measuring skill balance provides little support for the JOAT hypothesis that workers benefit from balanced skills. Balance does not predict entry once business skills and specialized skills are accounted for and has little relationship with success. The skills that matter in self-employment entry and success are business skills and specialized skills. Second, distinguishing basic from advanced self-employment reveals patterns that pooled analyses can hide. In particular, business skills are stronger predictors of entry and success in advanced self-employment, while specialized skills are important for success in basic self-employment. Third, the positive relationship between the occupation count and entry into basic self-employment partly reflects disrupted employment histories rather than balance over productive accumulation of skills.

Knowing which skills prepare workers for business ownership helps shape the advice and support available to aspiring owners. For workers planning to start a business, my results

suggest that the type of work experience matters more than its balance. Experience in occupations involving the management of people, budgets, and customers builds the business skills that predict entry into advanced self-employment and success thereafter, whereas experience across many unrelated jobs does not. Programs such as the U.S. Small Business Administration’s 7(j) Management and Technical Assistance and Emerging Leaders programs help prospective owners develop managerial and business skills. My findings are consistent with this approach rather than with encouraging workers to build balanced skills. Owners of basic businesses, who tend to earn less, show a different case. Many enter self-employment after job loss, and among them, specialized skills rather than business skills are associated with higher revenue and earnings. Support for these owners may therefore be more effective if it builds on their existing occupation-specific expertise rather than emphasizes general business training, and for some, displacement assistance may be more helpful than business training.

This analysis points to several directions for further work. The first is about interpretation. My estimates rely on selection on observables. The estimated marginal effects of business and specialized skills remain statistically significant as the NLSY79’s rich controls for wealth, risk tolerance, cognitive ability, personality, and family background are added, but tastes I do not observe may still shape both the skills a worker accumulates and her decision to enter self-employment. An instrument that shifts the skills a worker accumulates but affects self-employment only through those skills would address this endogeneity problem. A second is about whose skills matter. Owners can obtain skills they lack by forming a team, which I do not observe in the NLSY79. Data on founding teams would show whether the skills that prepare a successful business must be held by the owner or can be supplied by another member of the team. A third direction concerns skill measurement. My measures rely on occupation-level averages, which may mask differences in the tasks workers perform within the same occupation. Individual-level task data would capture this variation and provide a more direct measure of the skills workers accumulate on the job. A fourth is about

whether these skills can be built. Testing whether business skills developed through training program carry the same returns as those accumulated on the job would show how far preparation for business ownership can be acquired deliberately rather than over the course of a career.

The JOAT hypothesis asks how balanced an owner’s skills are. My findings suggest shifting the focus to which skills her business rewards. What prepares an owner for success is not skill balance, but the right type of skills for the type of business she runs.

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# Tables

Table 1: Distribution of Individuals by Number and Type of Self-Employment Entries

|                                                   | N. individuals | Share (%) |
|---------------------------------------------------|----------------|-----------|
| <i>Panel A: Number of Self-Employment Entries</i> |                |           |
| 0                                                 | 4,166          | 58.5      |
| 1                                                 | 1,838          | 25.8      |
| 2                                                 | 761            | 10.7      |
| 3                                                 | 240            | 3.4       |
| 4                                                 | 81             | 1.1       |
| 5                                                 | 20             | 0.3       |
| 6                                                 | 7              | 0.1       |
| 7                                                 | 1              | 0.0       |
| 8                                                 | 2              | 0.0       |
| Total                                             | 7,116          | 100.0     |
| <i>Panel B: Types of Self-Employment Entered</i>  |                |           |
| Never self-employed                               | 4,166          | 58.5      |
| Basic only                                        | 1,985          | 27.9      |
| Advanced only                                     | 596            | 8.4       |
| Both                                              | 369            | 5.2       |
| Total                                             | 7,116          | 100.0     |

Note: This table reports the number and share of individuals by the number of self-employment entries they are observed to make (Panel A) and by the types of self-employment they enter (Panel B). The unit of observation is the individual, and shares are calculated over all individuals in the entry sample. “Basic” and “Advanced” are defined as in Table 2; the categories in Panel B are mutually exclusive.

Table 2: Mean and Standard Deviations by Employment Type

|                                         | Wage     | Self-Employed |          |            | Diff.     |
|-----------------------------------------|----------|---------------|----------|------------|-----------|
|                                         | Employed | Any           | Basic    | Advanced   | (4)-(3)   |
|                                         | (1)      | (2)           | (3)      | (4)        | (5)       |
| <i>Panel A: Job Characteristics</i>     |          |               |          |            |           |
| Business revenue (excluding top-coded)  | .        | 119.40        | 55.11    | 212.27     | 157.16*** |
| (thousands of \$s)                      | (.)      | (311.16)      | (210.85) | (397.63)   |           |
| 1 if business revenue is top coded      | .        | 0.04          | 0.01     | 0.08       | 0.07***   |
|                                         | (.)      | (0.19)        | (0.09)   | (0.27)     |           |
| Weekly earnings (\$)                    | 528.53   | 530.05        | 426.37   | 851.65     | 425.28*** |
|                                         | (528.68) | (873.03)      | (672.73) | (1,259.80) |           |
| Hours worked per week                   | 37.59    | 33.80         | 31.42    | 41.22      | 9.79***   |
|                                         | (13.21)  | (20.60)       | (19.78)  | (21.34)    |           |
| 1 if job is ongoing (right-censored)    | 0.06     | 0.16          | 0.12     | 0.30       | 0.18***   |
|                                         | (0.24)   | (0.37)        | (0.32)   | (0.46)     |           |
| Job duration (years)                    | 3.22     | 4.40          | 3.54     | 7.09       | 3.56***   |
|                                         | (5.60)   | (5.98)        | (5.15)   | (7.42)     |           |
| 1 if job lasted less than 1 year        | 0.46     | 0.30          | 0.35     | 0.16       | -0.19***  |
|                                         | (0.50)   | (0.46)        | (0.48)   | (0.37)     |           |
| 1 if job lasted 5 years or more         | 0.17     | 0.27          | 0.21     | 0.47       | 0.26***   |
|                                         | (0.38)   | (0.44)        | (0.40)   | (0.50)     |           |
| Occupation                              |          |               |          |            |           |
| Management, business, science, and arts | 0.21     | 0.28          | 0.24     | 0.41       | 0.18***   |
| Service                                 | 0.22     | 0.29          | 0.35     | 0.11       | -0.24***  |
| Sales and office                        | 0.26     | 0.15          | 0.13     | 0.21       | 0.08***   |
| Natural resources, construction, and    | 0.11     | 0.19          | 0.20     | 0.16       | -0.04***  |
| maintenance                             |          |               |          |            |           |
| Production, transportation, and         | 0.19     | 0.09          | 0.08     | 0.11       | 0.03**    |
| material moving                         |          |               |          |            |           |
| Industry                                |          |               |          |            |           |
| Mining and construction                 | 0.08     | 0.16          | 0.15     | 0.16       | 0.01      |
| Manufacturing                           | 0.14     | 0.04          | 0.04     | 0.06       | 0.02***   |
| Transportation, communications, public  | 0.06     | 0.05          | 0.04     | 0.09       | 0.05***   |
| utilities                               |          |               |          |            |           |
| Wholesale and retail                    | 0.24     | 0.12          | 0.10     | 0.19       | 0.10***   |
| Services                                | 0.20     | 0.39          | 0.42     | 0.28       | -0.14***  |
| Professional services                   | 0.20     | 0.16          | 0.17     | 0.15       | -0.02     |
| Other                                   | 0.07     | 0.08          | 0.08     | 0.06       | -0.02**   |
| <i>Panel B: Work Experience</i>         |          |               |          |            |           |
| Potential experience (years)            | 12.05    | 16.44         | 15.80    | 18.45      | 2.65***   |
|                                         | (10.46)  | (10.60)       | (10.67)  | (10.13)    |           |
| Actual experience (years)               | 8.21     | 11.82         | 10.72    | 15.24      | 4.52***   |
|                                         | (8.93)   | (9.73)        | (9.46)   | (9.80)     |           |
| O*NET Business skills (years)           | 2.75     | 4.35          | 3.70     | 6.38       | 2.67***   |

Continued on next page

Table 2 (continued)

|                                                                           | Wage            | Self-Employed   |                 |                 | Diff.    |
|---------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|----------|
|                                                                           | Employed        | Any             | Basic           | Advanced        | (4)-(3)  |
|                                                                           | (1)             | (2)             | (3)             | (4)             | (5)      |
| O*NET Specialized skills (years)                                          | (3.75)<br>2.50  | (4.67)<br>3.89  | (4.18)<br>3.37  | (5.49)<br>5.52  | 2.15***  |
| O*NET Skill balance                                                       | (3.90)<br>28.37 | (4.61)<br>28.62 | (4.22)<br>28.55 | (5.32)<br>28.83 | 0.28***  |
| Occupation count (SOC major groups)<br>per year of actual experience      | (0.86)<br>0.78  | (0.78)<br>0.64  | (0.80)<br>0.70  | (0.69)<br>0.46  | -0.25*** |
| Number of involuntary job losses to date<br>per year of actual experience | (0.70)<br>0.37  | (0.62)<br>0.27  | (0.65)<br>0.30  | (0.49)<br>0.16  | -0.14*** |
| Number of past self-employment spells per<br>year of actual experience    | (0.50)<br>0.04  | (0.39)<br>0.06  | (0.42)<br>0.07  | (0.26)<br>0.06  | -0.01**  |
|                                                                           | (0.12)          | (0.12)          | (0.13)          | (0.11)          |          |
| <i>Panel C: Individual Characteristics</i>                                |                 |                 |                 |                 |          |
| 1 if female                                                               | 0.50<br>(0.50)  | 0.45<br>(0.50)  | 0.49<br>(0.50)  | 0.32<br>(0.47)  | -0.17*** |
| Race/Ethnicity                                                            |                 |                 |                 |                 |          |
| White                                                                     | 0.49            | 0.57            | 0.56            | 0.62            | 0.07***  |
| Hispanic                                                                  | 0.19            | 0.18            | 0.18            | 0.19            | 0.01     |
| Black                                                                     | 0.32            | 0.25            | 0.27            | 0.19            | -0.08*** |
| Marital status                                                            |                 |                 |                 |                 |          |
| Never married                                                             | 0.37            | 0.23            | 0.25            | 0.16            | -0.08*** |
| Cohabit                                                                   | 0.05            | 0.03            | 0.04            | 0.02            | -0.02*** |
| Married                                                                   | 0.39            | 0.54            | 0.51            | 0.64            | 0.13***  |
| Separated/Divorced/Widowed                                                | 0.19            | 0.20            | 0.20            | 0.18            | -0.02*   |
| 1 if child in household aged 6 or under                                   | 0.21<br>(0.41)  | 0.26<br>(0.44)  | 0.27<br>(0.44)  | 0.22<br>(0.42)  | -0.05*** |
| Risk tolerance                                                            |                 |                 |                 |                 |          |
| Least risk tolerant                                                       | 0.46            | 0.44            | 0.45            | 0.43            | -0.02    |
| Less risk tolerant                                                        | 0.12            | 0.13            | 0.14            | 0.13            | -0.01    |
| More risk tolerant                                                        | 0.15            | 0.15            | 0.15            | 0.16            | 0.02     |
| Most risk tolerant                                                        | 0.25            | 0.25            | 0.25            | 0.27            | 0.02     |
| Unknown risk tolerance                                                    | 0.02            | 0.02            | 0.02            | 0.01            | -0.01*** |
| Sociability personality trait at age 6                                    |                 |                 |                 |                 |          |
| Extremely shy                                                             | 0.15            | 0.13            | 0.14            | 0.11            | -0.04*** |
| Somewhat shy                                                              | 0.52            | 0.51            | 0.52            | 0.48            | -0.05*** |
| Somewhat outgoing                                                         | 0.22            | 0.23            | 0.22            | 0.27            | 0.05***  |
| Extremely outgoing                                                        | 0.11            | 0.12            | 0.11            | 0.15            | 0.03***  |
| Rotter Locus of Control (4-16)                                            | 7.50<br>(2.10)  | 7.34<br>(2.10)  | 7.40<br>(2.09)  | 7.18<br>(2.12)  | -0.21*** |
| Extraverted, enthusiastic (1-7)                                           | 5.13<br>(1.46)  | 5.16<br>(1.44)  | 5.12<br>(1.43)  | 5.26<br>(1.46)  | 0.14***  |
| Open to new experiences, complex (1-7)                                    | 5.29            | 5.34            | 5.32            | 5.39            | 0.07     |

Continued on next page

Table 2 (continued)

|                                        | Wage              | Self-Employed     |                   |                    | Diff.     |
|----------------------------------------|-------------------|-------------------|-------------------|--------------------|-----------|
|                                        | Employed          | Any               | Basic             | Advanced           | (4)-(3)   |
|                                        | (1)               | (2)               | (3)               | (4)                | (5)       |
| Conventional, uncreative (1-7)         | (1.34)<br>3.18    | (1.32)<br>3.06    | (1.31)<br>3.05    | (1.36)<br>3.07     | 0.02      |
| Net family wealth (thousands of \$s)   | (1.64)<br>79.40   | (1.64)<br>199.78  | (1.63)<br>158.31  | (1.67)<br>329.05   | 170.74*** |
| Net family income (thousands of \$s)   | (254.50)<br>52.04 | (570.48)<br>75.42 | (477.57)<br>64.42 | (779.34)<br>109.74 | 45.32***  |
| Education                              | (67.44)           | (122.71)          | (105.23)          | (161.07)           |           |
| Less than high school                  | 0.15              | 0.13              | 0.15              | 0.08               | -0.07***  |
| High school graduate                   | 0.47              | 0.43              | 0.45              | 0.38               | -0.08***  |
| Some college                           | 0.23              | 0.22              | 0.21              | 0.24               | 0.02      |
| College+                               | 0.15              | 0.22              | 0.19              | 0.31               | 0.12***   |
| AFQT percentile                        | 38.48             | 42.98             | 41.17             | 48.64              | 7.47***   |
|                                        | (28.30)           | (29.75)           | (29.61)           | (29.48)            |           |
| 1 if any family member owned business  | 0.27              | 0.34              | 0.32              | 0.40               | 0.08***   |
|                                        | (0.45)            | (0.47)            | (0.47)            | (0.49)             |           |
| 1 if considers oneself an entrepreneur | 0.19              | 0.38              | 0.33              | 0.52               | 0.20***   |
|                                        | (0.39)            | (0.48)            | (0.47)            | (0.50)             |           |
| N. spells                              | 60,071            | 4,566             | 3,457             | 1,109              |           |

*Note:* This table reports means at the start of each job spell, with standard deviations in parentheses for continuous variables. The sample includes every job that a worker starts while not already self-employed, so each self-employment spell corresponds to an entry into self-employment in the entry model. Rows for categorical variables report the share of spells in each category. “Basic” self-employment is unincorporated with start-up capital below 5,000 USD; “Advanced” is incorporated or has start-up capital of at least 5,000 USD. Revenue and earnings are in 2010 USD, adjusted with the regional CPI; revenue is available only for spells linked to the NLSY79 business ownership module. In Panel B, rows expressed per year of actual experience are computed only for spells with at least one year of actual experience. Column 5 reports the Advanced minus Basic difference, with significance from two-sample t-tests allowing unequal variances; for categorical variables the test is applied to an indicator for each category. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Table 3: Partial Correlation Coefficients of Experience Variables by Employment Type

|                                                     | (1)       | (2)       | (3)      | (4)      | (5)   |
|-----------------------------------------------------|-----------|-----------|----------|----------|-------|
| <i>Panel A: Wage Job Spells</i>                     |           |           |          |          |       |
| (1) O*NET Business skills                           | 1.000     |           |          |          |       |
| (2) O*NET Specialized skills                        | 0.431***  | 1.000     |          |          |       |
| (3) O*NET Skill balance                             | 0.107***  | -0.046*** | 1.000    |          |       |
| (4) Occupation count (SOC major groups)             | -0.161*** | -0.207*** | 0.294*** | 1.000    |       |
| (5) Number of involuntary job losses to date        | -0.245*** | -0.140*** | 0.144*** | 0.411*** | 1.000 |
| N. spells = 60,071                                  |           |           |          |          |       |
| <i>Panel B: Any Self-Employment Job Spells</i>      |           |           |          |          |       |
| (1) O*NET Business skills                           | 1.000     |           |          |          |       |
| (2) O*NET Specialized skills                        | 0.470***  | 1.000     |          |          |       |
| (3) O*NET Skill balance                             | 0.147***  | -0.009    | 1.000    |          |       |
| (4) Occupation count (SOC major groups)             | -0.174*** | -0.214*** | 0.244*** | 1.000    |       |
| (5) Number of involuntary job losses to date        | -0.265*** | -0.153*** | 0.120*** | 0.390*** | 1.000 |
| N. spells = 4,566                                   |           |           |          |          |       |
| <i>Panel C: Basic Self-Employment Job Spells</i>    |           |           |          |          |       |
| (1) O*NET Business skills                           | 1.000     |           |          |          |       |
| (2) O*NET Specialized skills                        | 0.398***  | 1.000     |          |          |       |
| (3) O*NET Skill balance                             | 0.154***  | -0.021    | 1.000    |          |       |
| (4) Occupation count (SOC major groups)             | -0.129*** | -0.176*** | 0.262*** | 1.000    |       |
| (5) Number of involuntary job losses to date        | -0.243*** | -0.119*** | 0.142*** | 0.375*** | 1.000 |
| N. spells = 3,457                                   |           |           |          |          |       |
| <i>Panel D: Advanced Self-Employment Job Spells</i> |           |           |          |          |       |
| (1) O*NET Business skills                           | 1.000     |           |          |          |       |
| (2) O*NET Specialized skills                        | 0.563***  | 1.000     |          |          |       |
| (3) O*NET Skill balance                             | 0.120***  | 0.003     | 1.000    |          |       |
| (4) Occupation count (SOC major groups)             | -0.228*** | -0.284*** | 0.225*** | 1.000    |       |
| (5) Number of involuntary job losses to date        | -0.288*** | -0.214*** | 0.084*** | 0.405*** | 1.000 |
| N. spells = 1,109                                   |           |           |          |          |       |

Note: This table reports partial correlation coefficients of experience measures at the start of each job spell, controlling for actual work experience, separately by employment type. The sample is the same as in Table 2. Business and specialized skills are time-in-occupation-weighted measures (in years). Skill balance is the effective number of skills (exponentiated Shannon index). Occupation count is the number of distinct SOC major groups held to date. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Table 4: Estimated Average Marginal Effects of Business, Specialized, and Balanced Skills on Self-Employment Entry Probabilities (Based on MNL and Logit Estimates Shown in Table A2)

|                                                                    | (1)                  | (2)                  | Dep. Var.: Type of Self-Employment Entry | (5)                  | (6)                 | (7)                  | (8)                  |
|--------------------------------------------------------------------|----------------------|----------------------|------------------------------------------|----------------------|---------------------|----------------------|----------------------|
| <i>Panel A: Basic SE (Unconditional Prob. = 0.014)</i>             |                      |                      |                                          |                      |                     |                      |                      |
| Std. O*NET Business skills                                         | -0.191***<br>(0.040) |                      | -0.135***<br>(0.041)                     | -0.142***<br>(0.043) |                     | -0.120***<br>(0.044) | -0.067<br>(0.045)    |
| Std. O*NET Specialized skills                                      |                      | -0.153***<br>(0.021) | -0.137***<br>(0.022)                     | -0.131***<br>(0.022) |                     | -0.109***<br>(0.023) | -0.115***<br>(0.023) |
| Std. O*NET Skill balance                                           |                      |                      |                                          | 0.013<br>(0.014)     | 0.017<br>(0.015)    | -0.010<br>(0.015)    | -0.013<br>(0.015)    |
| Std. N. occupation count                                           |                      |                      |                                          |                      | 0.148***<br>(0.013) | 0.128***<br>(0.015)  | 0.083***<br>(0.017)  |
| Std. N. involuntary job losses to date                             |                      |                      |                                          |                      |                     |                      | 0.113***<br>(0.020)  |
| <i>Panel B: Advanced SE (Unconditional Prob. = 0.004)</i>          |                      |                      |                                          |                      |                     |                      |                      |
| Std. O*NET Business skills                                         | 0.341***<br>(0.063)  |                      | 0.345***<br>(0.066)                      | 0.336***<br>(0.068)  |                     | 0.332***<br>(0.069)  | 0.354***<br>(0.070)  |
| Std. O*NET Specialized skills                                      |                      | 0.012<br>(0.032)     | -0.046<br>(0.033)                        | -0.045<br>(0.033)    |                     | -0.048<br>(0.033)    | -0.049<br>(0.033)    |
| Std. O*NET Skill balance                                           |                      |                      |                                          | 0.079***<br>(0.024)  | 0.013<br>(0.026)    | 0.017<br>(0.026)     | 0.016<br>(0.027)     |
| Std. N. occupation count                                           |                      |                      |                                          |                      | 0.012<br>(0.023)    | -0.014<br>(0.026)    | -0.036<br>(0.028)    |
| Std. N. involuntary job losses to date                             |                      |                      |                                          |                      |                     |                      | 0.064**<br>(0.030)   |
| <i>Panel C: Any SE, Binary Logit (Unconditional Prob. = 0.018)</i> |                      |                      |                                          |                      |                     |                      |                      |
| Std. O*NET Business skills                                         | -0.007<br>(0.034)    |                      | 0.036<br>(0.034)                         | 0.033<br>(0.036)     |                     | 0.052<br>(0.037)     | 0.097***<br>(0.038)  |
| Std. O*NET Specialized skills                                      |                      | -0.101***<br>(0.018) | -0.111***<br>(0.019)                     | -0.108***<br>(0.019) |                     | -0.093***<br>(0.019) | -0.098***<br>(0.019) |
| Std. O*NET Skill balance                                           |                      |                      |                                          | 0.028**<br>(0.012)   | 0.011<br>(0.013)    | -0.008<br>(0.013)    | -0.011<br>(0.013)    |
| Std. N. occupation count                                           |                      |                      |                                          |                      | 0.109***<br>(0.012) | 0.090***<br>(0.013)  | 0.051***<br>(0.015)  |
| Std. N. involuntary job losses to date                             |                      |                      |                                          |                      |                     |                      | 0.101***<br>(0.017)  |
| N. individuals                                                     | 7,116                | 7,116                | 7,116                                    | 7,116                | 7,116               | 7,116                | 7,116                |
| N. individual-year observations                                    | 251,214              | 251,214              | 251,214                                  | 251,214              | 251,214             | 251,214              | 251,214              |

Note: This table reports estimated average marginal effects from a multinomial logit on annual transitions into basic and advanced self-employment, on person-year observations for individuals not currently self-employed. The dependent variable takes three values: entry into basic SE, entry into advanced SE, or remaining in non-SE (reference). Panels A and B report estimates from the multinomial logit. Panel C reports estimates from a binary logit on entry into either type of self-employment, on the same sample and with the same controls. Marginal effects are computed for a 0.5 standard deviation increase in each skill, holding other variables at observed values, and expressed as a percentage change in the predicted probability, averaged over observations. All skill measures are standardized to mean zero and scaled so that a one-unit change equals half a standard deviation. Controls follow Section 4.5.5. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Table 5: Testing Selection on Observables: Estimated Average Marginal Effects on Self-Employment Entry (Based on MNL Estimates Shown in Table A3)

|                                                           | Dep. Var.: Type of Self-Employment Entry |                      |                      |                      |                      |                      |                      |                      |
|-----------------------------------------------------------|------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                           | (1)                                      | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  |
| <i>Panel A: Basic SE (Unconditional Prob. = 0.014)</i>    |                                          |                      |                      |                      |                      |                      |                      |                      |
| Std. O*NET Business skills                                | -0.065<br>(0.041)                        | -0.071*<br>(0.042)   | -0.105**<br>(0.042)  | -0.137***<br>(0.043) | -0.124***<br>(0.044) | -0.129***<br>(0.045) | -0.142***<br>(0.043) | -0.176***<br>(0.043) |
| Std. O*NET Specialized skills                             | -0.127***<br>(0.022)                     | -0.121***<br>(0.022) | -0.121***<br>(0.022) | -0.122***<br>(0.022) | -0.123***<br>(0.022) | -0.122***<br>(0.022) | -0.131***<br>(0.022) | -0.120***<br>(0.023) |
| Std. O*NET Skill balance                                  | 0.051***<br>(0.015)                      | 0.030*<br>(0.016)    | 0.028*<br>(0.016)    | 0.027*<br>(0.016)    | 0.026*<br>(0.016)    | 0.024<br>(0.016)     | 0.017<br>(0.015)     | 0.008<br>(0.015)     |
| <i>Panel B: Advanced SE (Unconditional Prob. = 0.004)</i> |                                          |                      |                      |                      |                      |                      |                      |                      |
| Std. O*NET Business skills                                | 0.419***<br>(0.064)                      | 0.484***<br>(0.065)  | 0.439***<br>(0.065)  | 0.353***<br>(0.066)  | 0.341***<br>(0.068)  | 0.337***<br>(0.069)  | 0.336***<br>(0.068)  | 0.228***<br>(0.067)  |
| Std. O*NET Specialized skills                             | -0.072**<br>(0.031)                      | -0.028<br>(0.032)    | -0.036<br>(0.032)    | -0.038<br>(0.032)    | -0.038<br>(0.032)    | -0.037<br>(0.032)    | -0.045<br>(0.033)    | -0.027<br>(0.034)    |
| Std. O*NET Skill balance                                  | 0.125***<br>(0.026)                      | 0.028<br>(0.027)     | 0.029<br>(0.026)     | 0.029<br>(0.026)     | 0.029<br>(0.026)     | 0.024<br>(0.026)     | 0.013<br>(0.026)     | 0.005<br>(0.026)     |
| Potential & actual experience                             | ✓                                        | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Demographics                                              |                                          | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Personality                                               |                                          |                      | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Resources                                                 |                                          |                      |                      | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Education & cognitive skills                              |                                          |                      |                      |                      | ✓                    | ✓                    | ✓                    | ✓                    |
| Family business background                                |                                          |                      |                      |                      |                      | ✓                    | ✓                    | ✓                    |
| N. past SE spells                                         |                                          |                      |                      |                      |                      |                      | ✓                    | ✓                    |
| Entrepreneur identity                                     |                                          |                      |                      |                      |                      |                      |                      | ✓                    |
| N. individuals                                            | 7,116                                    | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                |
| N. individual-year observations                           | 251,214                                  | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              |

Note: This table reports estimated average marginal effects of standardized business, specialized, and balanced skills on annual transitions into basic (Panel A) and advanced (Panel B) self-employment, across specifications that progressively add controls. Marginal effects are computed for a 0.5 standard deviation increase in each skill, holding other variables at observed values. All specifications control for potential and actual experience and their squares; columns 2–8 progressively add the control groups indicated at the bottom of the table. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

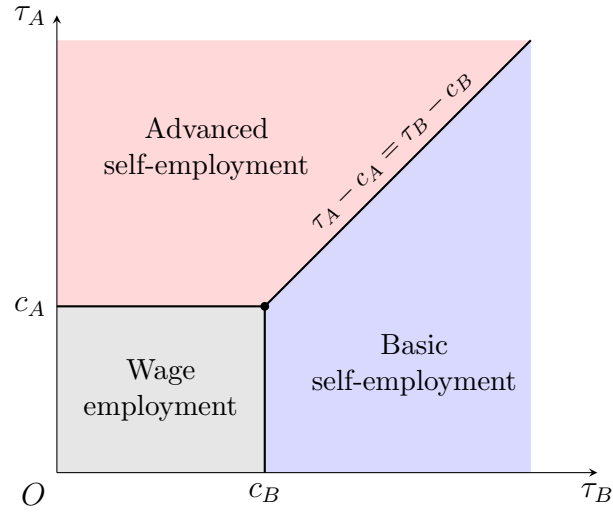
Table 6: Estimated Average Marginal Effects of Business, Specialized, and Balanced Skills on Self-Employment Revenue and Weekly Earnings (Based on Estimates Shown in Table A5)

|                               | Revenue Sample                |                              | Earnings Sample              |
|-------------------------------|-------------------------------|------------------------------|------------------------------|
|                               | Log Annual<br>Revenue (Tobit) | Log Weekly<br>Earnings (OLS) | Log Weekly<br>Earnings (OLS) |
|                               | (1)                           | (2)                          | (3)                          |
| <i>Panel A: Basic SE</i>      |                               |                              |                              |
| Std. O*NET Business skills    | 0.217*<br>(0.112)             | 0.092<br>(0.102)             | 0.022<br>(0.079)             |
| Std. O*NET Specialized skills | 0.173***<br>(0.048)           | 0.149***<br>(0.054)          | 0.192***<br>(0.043)          |
| Std. O*NET Skill balance      | 0.064<br>(0.042)              | 0.038<br>(0.043)             | 0.031**<br>(0.016)           |
| Tenure in SE job              | 0.152***<br>(0.014)           | 0.131***<br>(0.015)          | 0.097***<br>(0.008)          |
| N. observations               | 2,430                         | 2,430                        | 10,383                       |
| N. SE jobs                    | 1,221                         | 1,221                        | 4,118                        |
| Mean revenue or earnings      | 56,859                        | 412                          | 514                          |
| <i>Panel B: Advanced SE</i>   |                               |                              |                              |
| Std. O*NET Business skills    | 0.394***<br>(0.113)           | 0.142<br>(0.089)             | 0.163**<br>(0.076)           |
| Std. O*NET Specialized skills | 0.107<br>(0.068)              | 0.080<br>(0.059)             | 0.079<br>(0.049)             |
| Std. O*NET Skill balance      | -0.011<br>(0.065)             | 0.011<br>(0.047)             | 0.023<br>(0.030)             |
| Tenure in SE job              | 0.111***<br>(0.012)           | 0.104***<br>(0.010)          | 0.090***<br>(0.007)          |
| N. observations               | 2,465                         | 2,465                        | 5,938                        |
| N. SE jobs                    | 1,027                         | 1,027                        | 1,554                        |
| Mean revenue or earnings      | 215,578                       | 1,088                        | 1,088                        |

Note: This table reports estimated average marginal effects, estimated separately for basic (Panel A) and advanced (Panel B) self-employment, defined as in Table 2. Column 1 reports Tobit estimates for log annual revenue and column 2 reports OLS estimates for log weekly earnings, both on the revenue sample of self-employment job-years with reported revenue. Column 3 reports OLS estimates for log weekly earnings on the earnings sample of all self-employment job-years with reported earnings. Marginal effects are computed for a 0.5 standard deviation increase in each standardized skill, holding other variables at observed values; since the dependent variables are in logs, the estimates are interpretable as percentage changes. Controls follow Section 5.2. Mean revenue in column 1 excludes top-coded reports. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

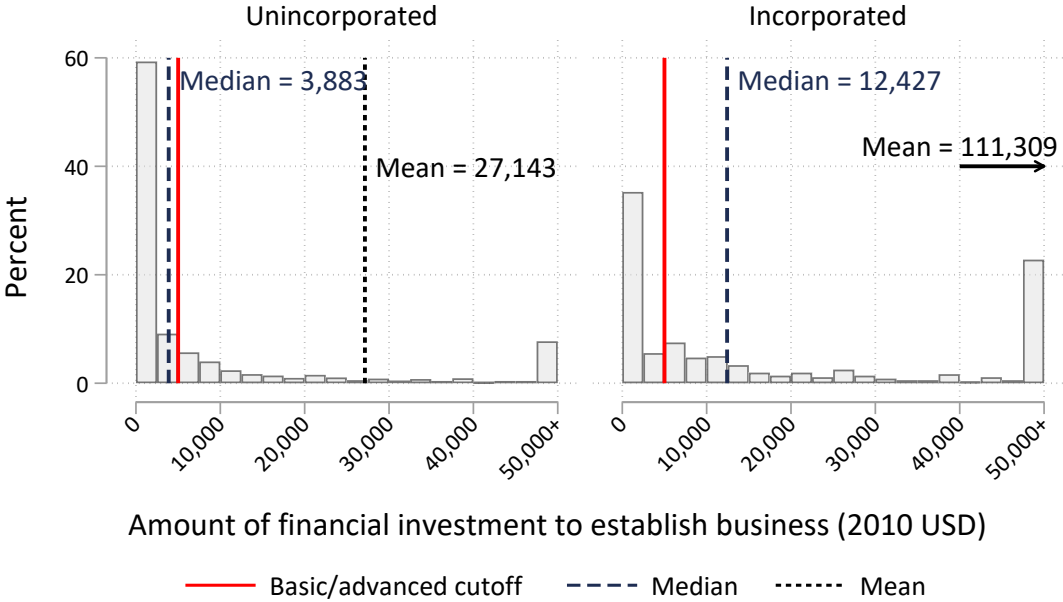
# Figures

Figure 1: The Worker's Choice for Each Pair of Tastes



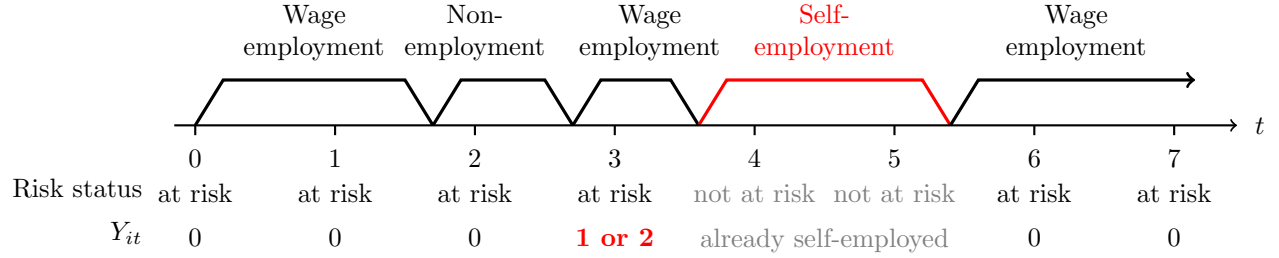
Note: This figure plots the option a worker chooses for each pair of tastes  $(\tau_B, \tau_A)$  for basic and advanced self-employment, given the opportunity costs  $c_B$  and  $c_A$ . Along the diagonal boundary, the worker is indifferent between the two types of self-employment. Her skills determine the opportunity costs and therefore the location of the boundaries, but not her tastes.

Figure 2: Distribution of the Amount of Start-Up Capital by Incorporation Status



Note: This figure plots the distribution of the amount of financial investment at the start of each self-employment spell, separately for incorporated and unincorporated spells. The solid line marks the cutoff between basic and advanced self-employment (5,000 USD), which is the median of positive amounts across all self-employment spells. Within each panel, the dashed lines mark the median and mean of positive amounts, and the arrow marks a mean that lies beyond the axis. Amounts of 50,000 USD or more are grouped in the last bar.

Figure 3: Illustrative Example of the Self-Employment Entry Outcome



Note: This figure plots the employment history of a hypothetical worker and the resulting entry outcome. Time  $t$  is measured in years since career start ( $t = 0$ ). In each year in which the worker is not self-employed, she is at risk of entry, and  $Y_{it}$  records her state over the next 12 months: 0 if she does not enter self-employment, 1 if she enters basic self-employment, and 2 if she enters advanced self-employment. Years spent in self-employment are not at risk and are excluded from the entry sample. After the self-employment spell ends, the worker is at risk again.

# Appendix

## Appendix A: Additional Tables

Table A1: Classification of Skill and Knowledge Elements in O\*NET Data by Skill Type

| Skill Type           | Skill Elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Knowledge Elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core Business</b> | Service Orientation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Economics and Accounting<br>Sales and Marketing<br>Customer and Personal Service<br>Production and Processing                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Managerial</b>    | Management of Financial Resources<br>Management of Material Resources<br>Management of Personnel Resources                                                                                                                                                                                                                                                                                                                                                                                                                              | Administration and Management<br>Personnel and Human Resources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Interpersonal</b> | Social Perceptiveness<br>Coordination<br>Persuasion<br>Negotiation<br>Instructing                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Other</b>         | Reading Comprehension<br>Active Listening<br>Writing<br>Speaking<br>Mathematics<br>Science<br>Critical Thinking<br>Active Learning<br>Learning Strategies<br>Monitoring<br>Complex Problem Solving<br>Operations Analysis<br>Technology Design<br>Equipment Selection<br>Installation<br>Programming<br>Operation Monitoring<br>Operation and Control<br>Equipment Maintenance<br>Troubleshooting<br>Repairing<br>Quality Control Analysis<br>Judgment and Decision Making<br>Systems Analysis<br>Systems Evaluation<br>Time Management | Clerical<br>Transportation<br>Food Production<br>Computers and Electronics<br>Engineering and Technology<br>Design<br>Building and Construction<br>Mechanical<br>Mathematics<br>Physics<br>Chemistry<br>Biology<br>Psychology<br>Sociology and Anthropology<br>Geography<br>Medicine and Dentistry<br>Therapy and Counseling<br>Education and Training<br>English Language<br>Foreign Language<br>Fine Arts<br>History and Archeology<br>Philosophy and Theology<br>Public Safety and Security<br>Law and Government<br>Telecommunications<br>Communications and Media |

Note: This table lists the O\*NET skill and knowledge elements assigned to each skill type used in constructing the business-skills measure.

Table A2: MNL and Logit Coefficient Estimates for Self-Employment Entry

|                                                           | Dep. Var.: Type of Self-Employment Entry |                      |                      |                     |                      |                      |                      |                      |
|-----------------------------------------------------------|------------------------------------------|----------------------|----------------------|---------------------|----------------------|----------------------|----------------------|----------------------|
|                                                           | (1)                                      | (2)                  | (3)                  | (4)                 | (5)                  | (6)                  | (7)                  | (8)                  |
| <i>Panel A: Basic SE (Unconditional Prob. = 0.014)</i>    |                                          |                      |                      |                     |                      |                      |                      |                      |
| Std. O*NET Business skills                                | -0.194***<br>(0.041)                     |                      | -0.137***<br>(0.042) |                     | -0.144***<br>(0.044) |                      | -0.121***<br>(0.045) | -0.067<br>(0.046)    |
| Std. O*NET Business skills <sup>2</sup>                   | 0.022***<br>(0.005)                      |                      | 0.028***<br>(0.006)  |                     | 0.028***<br>(0.006)  |                      | 0.025***<br>(0.006)  | 0.021***<br>(0.006)  |
| Std. O*NET Specialized skills                             |                                          | -0.154***<br>(0.022) | -0.138***<br>(0.023) |                     | -0.133***<br>(0.023) |                      | -0.110***<br>(0.024) | -0.116***<br>(0.024) |
| Std. O*NET Specialized skills <sup>2</sup>                |                                          | -0.003<br>(0.006)    | -0.012<br>(0.008)    |                     | -0.012<br>(0.007)    |                      | -0.011<br>(0.007)    | -0.012<br>(0.007)    |
| Std. O*NET Skill balance                                  |                                          |                      |                      | 0.013<br>(0.014)    | 0.017<br>(0.015)     |                      | -0.010<br>(0.015)    | -0.013<br>(0.015)    |
| Std. O*NET Skill balance <sup>2</sup>                     |                                          |                      |                      | -0.006*<br>(0.003)  | -0.004<br>(0.003)    |                      | -0.004<br>(0.003)    | -0.004<br>(0.003)    |
| Std. N. occupation count                                  |                                          |                      |                      |                     |                      | 0.151***<br>(0.014)  | 0.131***<br>(0.016)  | 0.085***<br>(0.018)  |
| Std. N. occupation count <sup>2</sup>                     |                                          |                      |                      |                     |                      | -0.011***<br>(0.004) | -0.010**<br>(0.004)  | -0.009**<br>(0.004)  |
| Std. N. involuntary job losses to date                    |                                          |                      |                      |                     |                      |                      |                      | 0.115***<br>(0.020)  |
| Std. N. involuntary job losses to date <sup>2</sup>       |                                          |                      |                      |                     |                      |                      |                      | -0.008**<br>(0.004)  |
| <i>Panel B: Advanced SE (Unconditional Prob. = 0.004)</i> |                                          |                      |                      |                     |                      |                      |                      |                      |
| Std. O*NET Business skills                                | 0.341***<br>(0.064)                      |                      | 0.344***<br>(0.066)  |                     | 0.336***<br>(0.069)  |                      | 0.332***<br>(0.069)  | 0.355***<br>(0.071)  |
| Std. O*NET Business skills <sup>2</sup>                   | -0.031***<br>(0.010)                     |                      | -0.013<br>(0.011)    |                     | -0.013<br>(0.011)    |                      | -0.013<br>(0.011)    | -0.014<br>(0.011)    |
| Std. O*NET Specialized skills                             |                                          | 0.011<br>(0.032)     | -0.047<br>(0.033)    |                     | -0.045<br>(0.033)    |                      | -0.047<br>(0.033)    | -0.049<br>(0.033)    |
| Std. O*NET Specialized skills <sup>2</sup>                |                                          | -0.043***<br>(0.010) | -0.048***<br>(0.012) |                     | -0.048***<br>(0.012) |                      | -0.048***<br>(0.012) | -0.048***<br>(0.012) |
| Std. O*NET Skill balance                                  |                                          |                      |                      | 0.079***<br>(0.024) | 0.014<br>(0.026)     |                      | 0.017<br>(0.027)     | 0.016<br>(0.027)     |
| Std. O*NET Skill balance <sup>2</sup>                     |                                          |                      |                      | 0.007<br>(0.005)    | -0.001<br>(0.007)    |                      | -0.001<br>(0.007)    | -0.001<br>(0.007)    |
| Std. N. occupation count                                  |                                          |                      |                      |                     |                      | 0.015<br>(0.023)     | -0.012<br>(0.026)    | -0.035<br>(0.028)    |
| Std. N. occupation count <sup>2</sup>                     |                                          |                      |                      |                     |                      | -0.003<br>(0.006)    | -0.001<br>(0.007)    | -0.001<br>(0.007)    |
| Std. N. involuntary job losses to date                    |                                          |                      |                      |                     |                      |                      |                      | 0.066**<br>(0.030)   |
| Std. N. involuntary job losses to date <sup>2</sup>       |                                          |                      |                      |                     |                      |                      |                      | -0.006<br>(0.004)    |
| N. individuals                                            | 7,116                                    | 7,116                | 7,116                | 7,116               | 7,116                | 7,116                | 7,116                | 7,116                |
| N. individual-year observations                           | 251,214                                  | 251,214              | 251,214              | 251,214             | 251,214              | 251,214              | 251,214              | 251,214              |

Table A2: MNL and Logit Coefficient Estimates for Self-Employment Entry (Continued)

|                                                                    | Dep. Var.: Type of Self-Employment Entry |                      |                      |                    |                      |                      |                      |                      |
|--------------------------------------------------------------------|------------------------------------------|----------------------|----------------------|--------------------|----------------------|----------------------|----------------------|----------------------|
|                                                                    | (1)                                      | (2)                  | (3)                  | (4)                | (5)                  | (6)                  | (7)                  | (8)                  |
| <i>Panel C: Any SE, Binary Logit (Unconditional Prob. = 0.018)</i> |                                          |                      |                      |                    |                      |                      |                      |                      |
| Std. O*NET Business skills                                         | -0.007<br>(0.034)                        |                      | 0.035<br>(0.035)     |                    | 0.032<br>(0.037)     |                      | 0.052<br>(0.038)     | 0.099**<br>(0.039)   |
| Std. O*NET Business skills <sup>2</sup>                            | 0.005<br>(0.005)                         |                      | 0.015***<br>(0.005)  |                    | 0.015***<br>(0.005)  |                      | 0.012**<br>(0.006)   | 0.009<br>(0.006)     |
| Std. O*NET Specialized skills                                      |                                          | -0.102***<br>(0.019) | -0.112***<br>(0.019) |                    | -0.109***<br>(0.020) |                      | -0.094***<br>(0.020) | -0.099***<br>(0.020) |
| Std. O*NET Specialized skills <sup>2</sup>                         |                                          | -0.014***<br>(0.005) | -0.023***<br>(0.006) |                    | -0.022***<br>(0.006) |                      | -0.022***<br>(0.006) | -0.022***<br>(0.006) |
| Std. O*NET Skill balance                                           |                                          |                      |                      | 0.028**<br>(0.012) | 0.011<br>(0.013)     |                      | -0.008<br>(0.013)    | -0.011<br>(0.013)    |
| Std. O*NET Skill balance <sup>2</sup>                              |                                          |                      |                      | -0.002<br>(0.003)  | -0.004<br>(0.003)    |                      | -0.004<br>(0.003)    | -0.004<br>(0.003)    |
| Std. N. occupation count                                           |                                          |                      |                      |                    |                      | 0.112***<br>(0.012)  | 0.093***<br>(0.014)  | 0.052***<br>(0.015)  |
| Std. N. occupation count <sup>2</sup>                              |                                          |                      |                      |                    |                      | -0.009***<br>(0.003) | -0.008**<br>(0.004)  | -0.007**<br>(0.004)  |
| Std. N. involuntary job losses to date                             |                                          |                      |                      |                    |                      |                      |                      | 0.103***<br>(0.018)  |
| Std. N. involuntary job losses to date <sup>2</sup>                |                                          |                      |                      |                    |                      |                      |                      | -0.008**<br>(0.003)  |
| N. individuals                                                     | 7,116                                    | 7,116                | 7,116                | 7,116              | 7,116                | 7,116                | 7,116                | 7,116                |
| N. individual-year observations                                    | 251,214                                  | 251,214              | 251,214              | 251,214            | 251,214              | 251,214              | 251,214              | 251,214              |

Note: This table reports coefficient estimates for the entry models reported in Table 4: multinomial logit coefficients in Panels A and B and binary logit coefficients in Panel C. Sample, dependent variable, and controls follow Table 4. Panels A and B report the log-odds of entering each type of self-employment relative to remaining in non-SE, while Panel C reports the log-odds of entering either type, so the coefficients in Panel C are not directly comparable to those in Panels A and B. All skill measures are standardized to mean zero and scaled so that a one-unit change equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Table A3: Testing Selection on Observables: MNL Coefficient Estimates for Self-Employment Entry

|                                                           | Dep. Var.: Type of Self-Employment Entry |                      |                      |                      |                      |                      |                      |                      |
|-----------------------------------------------------------|------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                           | (1)                                      | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  |
| <i>Panel A: Basic SE (Unconditional Prob. = 0.014)</i>    |                                          |                      |                      |                      |                      |                      |                      |                      |
| Std. O*NET Business skills                                | -0.066<br>(0.042)                        | -0.072*<br>(0.042)   | -0.106**<br>(0.043)  | -0.140***<br>(0.043) | -0.126***<br>(0.045) | -0.131***<br>(0.046) | -0.144***<br>(0.044) | -0.180***<br>(0.044) |
| Std. O*NET Business skills <sup>2</sup>                   | 0.025***<br>(0.006)                      | 0.026***<br>(0.006)  | 0.028***<br>(0.006)  | 0.031***<br>(0.006)  | 0.029***<br>(0.006)  | 0.031***<br>(0.006)  | 0.028***<br>(0.006)  | 0.029***<br>(0.006)  |
| Std. O*NET Specialized skills                             | -0.128***<br>(0.022)                     | -0.121***<br>(0.022) | -0.122***<br>(0.022) | -0.123***<br>(0.023) | -0.124***<br>(0.023) | -0.123***<br>(0.023) | -0.133***<br>(0.023) | -0.121***<br>(0.023) |
| Std. O*NET Specialized skills <sup>2</sup>                | -0.019**<br>(0.008)                      | -0.020***<br>(0.008) | -0.020***<br>(0.008) | -0.021***<br>(0.008) | -0.021***<br>(0.008) | -0.021***<br>(0.008) | -0.012<br>(0.007)    | -0.012*<br>(0.007)   |
| Std. O*NET Skill balance                                  | 0.053***<br>(0.015)                      | 0.031*<br>(0.016)    | 0.029*<br>(0.016)    | 0.028*<br>(0.016)    | 0.027*<br>(0.016)    | 0.024<br>(0.016)     | 0.017<br>(0.015)     | 0.008<br>(0.015)     |
| Std. O*NET Skill balance <sup>2</sup>                     | -0.002<br>(0.003)                        | -0.004<br>(0.003)    | -0.004<br>(0.003)    | -0.004<br>(0.003)    | -0.004<br>(0.003)    | -0.004<br>(0.003)    | -0.004<br>(0.003)    | -0.005<br>(0.003)    |
| <i>Panel B: Advanced SE (Unconditional Prob. = 0.004)</i> |                                          |                      |                      |                      |                      |                      |                      |                      |
| Std. O*NET Business skills                                | 0.420***<br>(0.065)                      | 0.486***<br>(0.065)  | 0.440***<br>(0.066)  | 0.353***<br>(0.066)  | 0.341***<br>(0.069)  | 0.337***<br>(0.069)  | 0.336***<br>(0.069)  | 0.227***<br>(0.067)  |
| Std. O*NET Business skills <sup>2</sup>                   | -0.014<br>(0.011)                        | -0.023**<br>(0.011)  | -0.019*<br>(0.011)   | -0.015<br>(0.011)    | -0.013<br>(0.011)    | -0.012<br>(0.011)    | -0.013<br>(0.011)    | -0.008<br>(0.011)    |
| Std. O*NET Specialized skills                             | -0.072**<br>(0.032)                      | -0.027<br>(0.032)    | -0.035<br>(0.032)    | -0.038<br>(0.032)    | -0.037<br>(0.032)    | -0.036<br>(0.032)    | -0.045<br>(0.033)    | -0.026<br>(0.035)    |
| Std. O*NET Specialized skills <sup>2</sup>                | -0.046***<br>(0.011)                     | -0.054***<br>(0.012) | -0.052***<br>(0.012) | -0.055***<br>(0.012) | -0.055***<br>(0.012) | -0.055***<br>(0.012) | -0.048***<br>(0.012) | -0.046***<br>(0.011) |
| Std. O*NET Skill balance                                  | 0.126***<br>(0.026)                      | 0.029<br>(0.027)     | 0.029<br>(0.026)     | 0.029<br>(0.026)     | 0.030<br>(0.026)     | 0.024<br>(0.026)     | 0.014<br>(0.026)     | 0.005<br>(0.026)     |
| Std. O*NET Skill balance <sup>2</sup>                     | 0.009*<br>(0.005)                        | -0.001<br>(0.007)    | -0.001<br>(0.007)    | -0.000<br>(0.007)    | -0.000<br>(0.007)    | -0.000<br>(0.007)    | -0.001<br>(0.007)    | -0.001<br>(0.007)    |
| Potential & actual experience                             | ✓                                        | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Demographics                                              |                                          | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Personality                                               |                                          |                      | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Resources                                                 |                                          |                      |                      | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Education & cognitive skills                              |                                          |                      |                      |                      | ✓                    | ✓                    | ✓                    | ✓                    |
| Family business background                                |                                          |                      |                      |                      |                      | ✓                    | ✓                    | ✓                    |
| N. past SE spells                                         |                                          |                      |                      |                      |                      |                      | ✓                    | ✓                    |
| Entrepreneur identity                                     |                                          |                      |                      |                      |                      |                      |                      | ✓                    |
| N. individuals                                            | 7,116                                    | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                |
| N. individual-year observations                           | 251,214                                  | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              |

Note: This table reports MNL coefficient estimates underlying the selection-on-observables analysis in Table 5. Sample, dependent variable, and progressive controls follow Table 5. All skill measures are standardized to mean zero and scaled so that a one-unit change equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Table A4: Are Business Skills Reflecting Experience in High-Skill Occupations?

|                                                           | Dep. Var.: Type of Self-Employment Entry |                      |                      |                      |                      |                      |
|-----------------------------------------------------------|------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                           | (1)                                      | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
| <i>Panel A: Basic SE (Unconditional Prob. = 0.014)</i>    |                                          |                      |                      |                      |                      |                      |
| Std. Actual exp. in Job Zone 4+ occupations               | -0.063**<br>(0.025)                      | -0.013<br>(0.028)    | -0.044*<br>(0.026)   | -0.018<br>(0.028)    | -0.068***<br>(0.026) | -0.020<br>(0.029)    |
| Std. O*NET Business skills                                |                                          | -0.183***<br>(0.044) |                      | -0.127***<br>(0.045) |                      | -0.134***<br>(0.046) |
| Std. O*NET Specialized skills                             |                                          |                      | -0.144***<br>(0.022) | -0.132***<br>(0.022) |                      | -0.126***<br>(0.023) |
| Std. O*NET Skill balance                                  |                                          |                      |                      |                      | 0.020<br>(0.014)     | 0.017<br>(0.015)     |
| <i>Panel B: Advanced SE (Unconditional Prob. = 0.004)</i> |                                          |                      |                      |                      |                      |                      |
| Std. Actual exp. in Job Zone 4+ occupations               | 0.089**<br>(0.035)                       | 0.003<br>(0.038)     | 0.082**<br>(0.036)   | -0.010<br>(0.039)    | 0.064*<br>(0.036)    | -0.013<br>(0.039)    |
| Std. O*NET Business skills                                |                                          | 0.334***<br>(0.068)  |                      | 0.350***<br>(0.071)  |                      | 0.342***<br>(0.072)  |
| Std. O*NET Specialized skills                             |                                          |                      | -0.001<br>(0.033)    | -0.046<br>(0.033)    |                      | -0.045<br>(0.033)    |
| Std. O*NET Skill balance                                  |                                          |                      |                      |                      | 0.069***<br>(0.025)  | 0.014<br>(0.026)     |
| N. individuals                                            | 7,116                                    | 7,116                | 7,116                | 7,116                | 7,116                | 7,116                |
| N. individual-year observations                           | 251,214                                  | 251,214              | 251,214              | 251,214              | 251,214              | 251,214              |

Note: This table tests whether the estimated effects of business skills in Table 4 reflect business-specific skill content rather than a general high-skill occupation effect, by adding controls for O\*NET job zone (a five-level classification of the preparation required for an occupation). Sample, dependent variable, and other controls follow Table 4. All skill measures are standardized to mean zero and scaled so that a one-unit change equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Table A5: Tobit and OLS Coefficient Estimates for Self-Employment Revenue and Weekly Earnings

|                                                       | Revenue Sample                |                              | Earnings Sample              |
|-------------------------------------------------------|-------------------------------|------------------------------|------------------------------|
|                                                       | Log Annual<br>Revenue (Tobit) | Log Weekly<br>Earnings (OLS) | Log Weekly<br>Earnings (OLS) |
|                                                       | (1)                           | (2)                          | (3)                          |
| <i>Panel A: Basic SE</i>                              |                               |                              |                              |
| Std. O*NET Business skills                            | 0.208*<br>(0.107)             | 0.081<br>(0.098)             | 0.012<br>(0.062)             |
| Std. O*NET Business skills <sup>2</sup>               | -0.023<br>(0.016)             | -0.026*<br>(0.014)           | -0.004<br>(0.011)            |
| Std. O*NET Specialized skills                         | 0.169***<br>(0.047)           | 0.150***<br>(0.052)          | 0.182***<br>(0.035)          |
| Std. O*NET Specialized skills <sup>2</sup>            | -0.010<br>(0.012)             | 0.001<br>(0.011)             | -0.005<br>(0.011)            |
| Std. O*NET Skill balance                              | 0.072<br>(0.044)              | 0.043<br>(0.046)             | 0.034*<br>(0.018)            |
| Std. O*NET Skill balance <sup>2</sup>                 | 0.013**<br>(0.006)            | 0.007<br>(0.006)             | 0.002<br>(0.002)             |
| Tenure in SE job                                      | 0.194***<br>(0.020)           | 0.167***<br>(0.021)          | 0.113***<br>(0.010)          |
| Tenure in SE job <sup>2</sup>                         | -0.004***<br>(0.001)          | -0.003***<br>(0.001)         | -0.003***<br>(0.000)         |
| N. observations                                       | 2,430                         | 2,430                        | 10,383                       |
| N. SE jobs                                            | 1,221                         | 1,221                        | 4,118                        |
| Adj. R <sup>2</sup> (pseudo R <sup>2</sup> for Tobit) | 0.057                         | 0.166                        | 0.183                        |
| Mean revenue or earnings                              | 56,859                        | 412                          | 514                          |
| <i>Panel B: Advanced SE</i>                           |                               |                              |                              |
| Std. O*NET Business skills                            | 0.413***<br>(0.118)           | 0.156*<br>(0.092)            | 0.130*<br>(0.070)            |
| Std. O*NET Business skills <sup>2</sup>               | -0.049***<br>(0.017)          | -0.035**<br>(0.015)          | -0.034***<br>(0.012)         |
| Std. O*NET Specialized skills                         | 0.106<br>(0.071)              | 0.071<br>(0.061)             | 0.094**<br>(0.044)           |
| Std. O*NET Specialized skills <sup>2</sup>            | 0.003<br>(0.014)              | 0.026**<br>(0.012)           | 0.020**<br>(0.010)           |
| Std. O*NET Skill balance                              | -0.016<br>(0.060)             | 0.008<br>(0.043)             | 0.022<br>(0.030)             |
| Std. O*NET Skill balance <sup>2</sup>                 | 0.008<br>(0.010)              | 0.005<br>(0.007)             | 0.002<br>(0.004)             |
| Tenure in SE job                                      | 0.182***<br>(0.023)           | 0.195***<br>(0.019)          | 0.135***<br>(0.011)          |
| Tenure in SE job <sup>2</sup>                         | -0.003***<br>(0.001)          | -0.004***<br>(0.001)         | -0.003***<br>(0.000)         |
| N. observations                                       | 2,465                         | 2,465                        | 5,938                        |
| N. SE jobs                                            | 1,027                         | 1,027                        | 1,554                        |
| Adj. R <sup>2</sup> (pseudo R <sup>2</sup> for Tobit) | 0.057                         | 0.164                        | 0.158                        |
| Mean revenue or earnings                              | 215,578                       | 1,088                        | 1,088                        |

Note: This table reports the Tobit (column 1) and OLS (columns 2–3) coefficient estimates underlying the marginal effects in Table 6. Samples, dependent variables, and controls follow Table 6. Each skill measure is standardized with the mean and standard deviation of the revenue sample and scaled so that a one-unit change equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

## Appendix B: Other Approaches to the Endogeneity Problem

The estimates in Section 5.3 rely on selection on observables. Two other approaches could address the endogeneity problem without requiring me to observe the tastes that create it: individual fixed effects and an instrument for skills. In this appendix, I describe what each approach would address and why neither can be used here.

The error term  $\epsilon_{ijt}$  in equation (11) captures factors that shift the utility of option  $j$  but are not included in the observed regressors. Because  $BX_{it}$  and  $SX_{it}$  are built from workers' occupational histories, they reflect not only the skills of the occupations workers hold but also the choices that determine which occupations they enter and stay in. The baseline specification controls for experience, demographic characteristics, wealth, risk tolerance, cognitive ability, personality traits, and family background, but tastes I do not observe may still shape how workers allocate their years across occupations.

To show how this may affect the estimates, decompose  $\epsilon_{ijt}$  as  $\epsilon_{ijt} = u_{ij} + v_{ijt}$ , where  $u_{ij}$  is time-invariant and  $v_{ijt}$  varies within a worker across years. The time-invariant component  $u_{ij}$  captures tastes that do not change over a worker's career, such as a desire for autonomy or an aspiration to own a business. A worker with such tastes may sort into occupations that use more business skills and stay in them longer in anticipation of entry. This creates a positive correlation between  $BX_{it}$  and  $u_{ij}$  for the self-employment options, so the estimated association between business skills and entry may reflect both the skills themselves and these tastes.

The time-varying component  $v_{ijt}$  captures changes in a worker's plans or circumstances over the career. For example, a worker who becomes interested in self-employment in mid-career may move into an occupation that uses more business skills before entering. In this case, within-person changes in skills are also correlated with unobserved changes in the attractiveness of self-employment. This distinction parallels work on returns to occupational tenure (Kambourov and Manovskii, 2009) and task-specific human capital (Gathmann and Schönberg, 2010), where workers sort into occupations or tasks based on match quality.

## B.1 Individual Fixed Effects

Individual fixed effects would address the time-invariant component  $u_{ij}$ . The relationship between skills and entry would then be identified from within-person changes in  $BX_{it}$  and  $SX_{it}$  as workers move across occupations, so tastes that do not change over a career would no longer drive the estimates. Because the entry model is a logit, this would require a conditional logit model following Chamberlain (1980). This approach avoids imposing a distributional assumption on the individual effects and allows  $u_{ij}$  to be arbitrarily correlated with the observed covariates. A conditional rather than unconditional approach is necessary because estimating individual-specific intercepts in a nonlinear model introduces the incidental parameters problem.

Fixed effects, however, would not address the time-varying component  $v_{ijt}$ . A worker who becomes interested in self-employment in mid-career and then moves into occupations that use more business skills would still generate within-person variation in  $BX_{it}$  that is correlated with her changing plans.

Fixed effects are also of limited use for entry. Business and specialized skills accumulate steadily as a worker gains experience, so once experience is controlled for, little within-person variation in skills remains to identify their association with entry. Fixed effects are more informative for earnings, because there the return to skills in self-employment does not have to come from changes in a worker's skills. It comes from how much a worker's earnings change when she moves between wage employment and self-employment, and from how that change differs with the skills she brings. The fixed effect absorbs her overall earnings level, while the move itself provides the within-person variation. Appendix C uses this approach.

## B.2 Instrumental Variables

An instrument would address both components of the error term. It would need to shift the skills a worker accumulates but affect self-employment only through those skills. One candidate is the industrial composition of the county where a worker starts her career, in the

spirit of Jara-Figueroa et al. (2018). Local labor markets differ in the occupations they offer. A county whose economy is concentrated in manufacturing exposes workers to a different set of occupations, and therefore a different path of skill accumulation, than a county with a larger service sector or a concentration of professional industries. Workers with similar initial characteristics may thus accumulate different skills because they begin their careers in different local labor markets.

I construct such an instrument for business skills. For each county  $c$ , I use the 1975 County Business Patterns to calculate the share  $s_{kc}$  of employment in each industry  $k$ . I then measure how much each industry uses business skills,  $\overline{BX}_k$ , by assigning each occupation its business-skill score from Section 4.5 and averaging across workers employed in industry  $k$  in the 1980 Census. The instrument for worker  $i$ , whose county of residence at labor market entry is  $c(i)$ , is:

$$Z_{ict}^{BX} = \sum_k s_{kc(i),1975} \cdot g_{kt} \cdot \overline{BX}_k, \quad (\text{B1})$$

where  $g_{kt}$  is national cumulative employment growth in industry  $k$  from 1975 to year  $t$ . The instrument is higher for workers who started their careers in counties where industries that use more business skills accounted for a large share of employment and later grew nationally.

I construct a parallel instrument for specialized skills:

$$Z_{ict}^{SX} = \sum_k s_{kc(i),1975} \cdot \overline{SX}_{kt}. \quad (\text{B2})$$

Because specialized skills depend on each worker's target occupation, the industry score  $\overline{SX}_{kt}$  cannot simply average occupation-level scores. Instead, it measures how much the occupations within industry  $k$  share their most important skills. In construction, for example, plumbers, electricians, and carpenters rely on many of the same skills, so working in any of these occupations builds skills that are useful in the others. In hospitals, nurses, accountants,

and janitors share few of their most important skills. Workers who start their careers in counties dominated by industries of the first kind accumulate more transferable specialized skills. The score changes over time as the occupational mix within each industry changes, which I measure at each Census and American Community Survey benchmark from 1980 to 2015.

For either instrument to be valid, the industrial composition of a worker's starting county must affect self-employment only through the skills she accumulates. Two concerns remain. First, local economic conditions may directly affect self-employment decisions. For example, limited wage opportunities may push workers into self-employment regardless of their skills. Controlling for the county unemployment rate at labor market entry would absorb part of this channel. Second, workers may sort across counties based on their tastes for self-employment. Using the county of residence at labor market entry, rather than at the time of the self-employment decision, reduces this concern, as does restricting the sample to workers who stay in the same county during the period used to construct the instrument.

Two problems prevent me from using these instruments. First, they are highly correlated: across counties, their correlation is about 0.9. Industries whose occupations share business skills, such as finance and professional services, also rank high on how much their occupations share specialized skills. The instruments therefore cannot separate the two types of skills, which the entry model requires. Second, the business-skill instrument is weak in a first stage with individual fixed effects, the setting of Appendix C, where its only source of variation within a worker is national industry growth over time. Separating business from specialized skills would require a source of variation that shifts one type of skill without shifting the other, which I leave for future work.

## Appendix C: Earnings Analysis Using Individual Fixed Effects

Revenue measures the performance of the business, but it is observed only for workers who report operating a business, and it cannot be compared with pay in wage employment. I therefore also examine weekly earnings, which are observed both before and after entry. Earnings are an imperfect measure of business performance, because owners choose how much income to withdraw from the business. They may retain profits within the firm, pay themselves a relatively small salary, or reinvest earnings to support future growth. Following the same workers across wage employment and self-employment, however, shows how skills shape the earnings path over a spell.

The earnings FE sample is used to estimate the wage models described below, which compare earnings across wage employment and self-employment among individuals who enter self-employment. I restrict the entry sample to the 2,503 individuals who enter self-employment at least once during the observation period.

The unit of observation is a person-interview-job, and the sample includes wage observations from both non-self-employment and self-employment periods. For non-self-employed periods, I restrict attention to wage employment because earnings are only observed for individuals working in wage jobs. During self-employment spells, I include observations when self-employment earnings are reported for the relevant self-employment job. Each individual contributes one observation for each job and interview period in which earnings are available. Because earnings are reported at the interview date, variables are measured at the interview level rather than over a twelve-month window. The final sample contains 69,586 observations for individuals who enter basic self-employment and 32,096 observations for individuals who enter advanced self-employment.

I examine weekly earnings using an individual fixed-effects specification that lets the

returns to each skill vary over the self-employment spell:

$$\ln w_{it} = \alpha_i + \sum_{k=0}^3 (\delta_k + \gamma_k BX_{it} + \theta_k SX_{it} + \pi_k BSC_{it}) D_{it}^{(k)} + X'_{it} \beta + \epsilon_{it}, \quad (C1)$$

where  $D_{it}^{(0)} = SE_{it}$ ,  $D_{it}^{(1)} = 1stYearSE_{it}$ ,  $D_{it}^{(2)} = SE_{it}\tau_{it}$ , and  $D_{it}^{(3)} = SE_{it}\tau_{it}^2$ , and  $BX_{it}$ ,  $SX_{it}$ , and  $BSC_{it}$  are as defined above.

The dependent variable  $\ln w_{it}$  is log weekly earnings in 2010 USD, and the individual fixed effect  $\alpha_i$  absorbs all time-invariant individual characteristics.  $SE_{it}$  equals one when the worker is in the self-employment type of interest and zero in wage employment.  $1stYearSE_{it}$  is an indicator for the first year in self-employment.  $\tau_{it}$  is tenure in the current self-employment job, set to zero in the first year of self-employment and outside self-employment. Together, these four terms trace the earnings path over a spell:  $\delta_0$  and  $\delta_1$  give the gap relative to wage employment at entry, and  $\delta_2$  and  $\delta_3$  the subsequent growth with tenure. Interacting each of them with business skills, specialized skills, and skill balance allows each skill to shift both the initial gap and the growth profile. The control vector  $X_{it}$  includes quadratics in actual and potential experience, cohabitation and marital status, whether the age of the youngest child is 6 or younger, and region and industry fixed effects, together with actual experience at self-employment entry interacted with  $1stYearSE_{it}$  to control for the timing of the transition. I estimate the specification separately for workers who ever enter basic and advanced self-employment. Standard errors are clustered at the individual level. To address survivorship bias, I weight observations by the inverse of the number of observations within each job spell, so that each spell receives equal weight regardless of length. I first estimate a simpler version of equation (C1) that omits the skill interactions, which establishes the baseline trajectory against which the skill measures shift earnings.

Before introducing the skill interactions, I estimate a simpler version of equation (C1) in

which the four self-employment terms enter on their own, setting  $\gamma_k = \theta_k = \pi_k = 0$  for all  $k$ :

$$\ln w_{it} = \alpha_i + \sum_{k=0}^3 \delta_k D_{it}^{(k)} + X'_{it} \beta + \epsilon_{it}. \quad (\text{C2})$$

All terms are defined as above, and the specification is estimated on the same sample of workers who have ever entered self-employment.

I estimate equation (C2) separately for workers who ever enter basic and advanced self-employment. Table C1 reports the estimates.  $\delta_0$  captures the earnings gap between self-employment and wage employment after the first year of self-employment (when  $\tau_{it} = 0$  and  $\text{1stYearSE}_{it} = 0$ ), and  $\delta_1$  captures the additional adjustment in the first year; the total gap at entry is  $\delta_0 + \delta_1$ , and a negative value is consistent with an initial earnings drop.  $\delta_2$  and  $\delta_3$  trace the subsequent earnings growth over self-employment tenure.

The estimates show two patterns. First, entering self-employment is accompanied by an immediate earnings drop relative to wage employment, which is more pronounced for basic self-employment. In basic self-employment, the first-year earnings gap is about 43 log points ( $\delta_0 + \delta_1 = -0.227 - 0.199$ ). In advanced self-employment, the first-year earnings gap is about 31 log points ( $\delta_0 + \delta_1 = -0.430 + 0.116$ , with  $\delta_1$  not statistically significant). Second, earnings grow faster in self-employment than in the counterfactual wage path. The tenure return  $\delta_2$  implies annual earnings growth of about 6 log points in basic self-employment and 11 log points in advanced self-employment, on top of the potential and actual experience profile. Combined with the initial drop, this produces a U-shaped trajectory in which earnings dip at entry and then recover with tenure, more sharply in advanced self-employment. The analysis below reads the skill-specific earnings paths against this baseline.

To illustrate how skills shape earnings over the self-employment spell, I generate predicted earnings paths for wage employment, self-employment, and the counterfactual wage employment path. Most covariates are held at their observed values. For the experience measures, I assume workers enter self-employment after 10 years of actual experience and

remain continuously employed, so potential and actual experience are equal throughout the spell. I then vary each skill measure from its mean to one standard deviation above the mean. Figures C1 and C2 plot predicted log weekly earnings against actual experience for basic and advanced self-employment, separately by business skills, specialized skills, and skill balance.

Business skills shape the earnings path differently in the two types of self-employment. Figures C1 and C2 plot predicted earnings before and after entry into self-employment, alongside the wage path those workers would have followed had they not entered. The baseline estimates above give the average pattern, a drop at entry followed by faster growth with tenure. The figures show whether that drop and that growth differ with business skills. In basic self-employment, workers with higher business skills experience a larger earnings decline at entry but recover somewhat faster. Their earnings remain above those of workers with lower business skills but never catch up to the wage path. In advanced self-employment, higher business skills are associated with a much larger and steadily growing earnings premium. Earnings eventually catch up to and exceed the counterfactual wage path, and the gap continues to widen with tenure. One interpretation is that the returns to business skills accumulate as the business develops.

Specialized skills raise earnings among those who enter self-employment. The returns are largest in basic self-employment, where workers with higher specialized skills earn substantially more throughout the spell. In advanced self-employment, specialized skills mainly reduce the initial earnings loss at entry, and the gap between workers with high and low specialized skills narrows over time as the two paths converge. It is possible that specialized skills are productive from entry, rather than becoming more valuable as the business develops.

Workers with more balanced skills earn less in both types of self-employment, and the gap widens with tenure. The revenue sample shows no such association (Table 6), and the two estimates come from different samples and specifications, so I do not read the negative

result as robust.

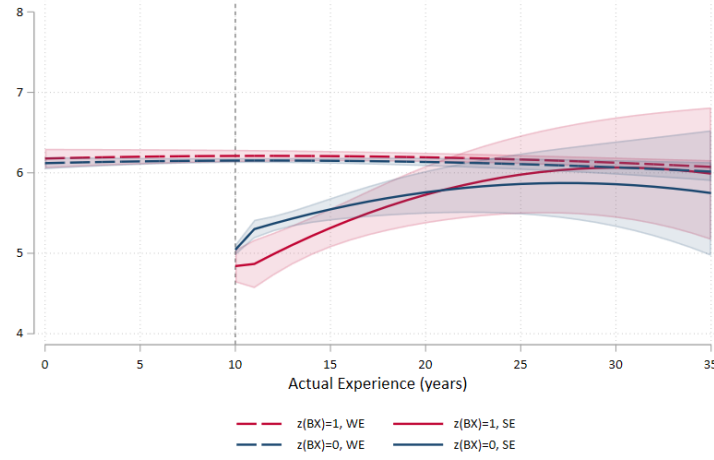
Table C1: Baseline Coefficient Estimates for Log Earnings with Individual Fixed Effects

|                                                  | Dep. Var.: Log(Weekly Earnings + 1) |                      |                      |                      |
|--------------------------------------------------|-------------------------------------|----------------------|----------------------|----------------------|
|                                                  | Basic SE                            |                      | Advanced SE          |                      |
|                                                  | (1)                                 | (2)                  | (3)                  | (4)                  |
| Actual experience                                | 0.063***<br>(0.005)                 | 0.050***<br>(0.005)  | 0.069***<br>(0.010)  | 0.060***<br>(0.010)  |
| Actual experience <sup>2</sup>                   | -0.001***<br>(0.000)                | -0.000***<br>(0.000) | -0.001***<br>(0.000) | -0.001***<br>(0.000) |
| Potential experience                             | -0.027***<br>(0.005)                | -0.019***<br>(0.005) | -0.029***<br>(0.010) | -0.026***<br>(0.009) |
| Potential experience <sup>2</sup>                | 0.000<br>(0.000)                    | -0.000<br>(0.000)    | 0.000<br>(0.000)     | 0.000<br>(0.000)     |
| 1 if basic self-employment                       | -0.482***<br>(0.071)                | -0.432***<br>(0.070) |                      |                      |
| 1 if advanced self-employment                    |                                     |                      | -0.421***<br>(0.099) | -0.412***<br>(0.109) |
| 1 if first year in self-employment               | 0.111<br>(0.071)                    | 0.055<br>(0.075)     | 0.275**<br>(0.117)   | 0.141<br>(0.146)     |
| Tenure in self-employment job - 1                | 0.032**<br>(0.013)                  | 0.047***<br>(0.016)  | 0.069***<br>(0.012)  | 0.095***<br>(0.016)  |
| (Tenure in self-employment job - 1) <sup>2</sup> | -0.001<br>(0.000)                   | -0.001<br>(0.001)    | -0.002***<br>(0.000) | -0.002***<br>(0.000) |
| Weighted by inverse of spell-report frequency    |                                     | ✓                    |                      | ✓                    |
| Mean dep. var.                                   | 5.998                               | 5.998                | 6.299                | 6.299                |
| N. obs.                                          | 69,586                              | 69,586               | 32,096               | 32,096               |
| Adjusted R <sup>2</sup>                          | 0.376                               | 0.354                | 0.290                | 0.298                |

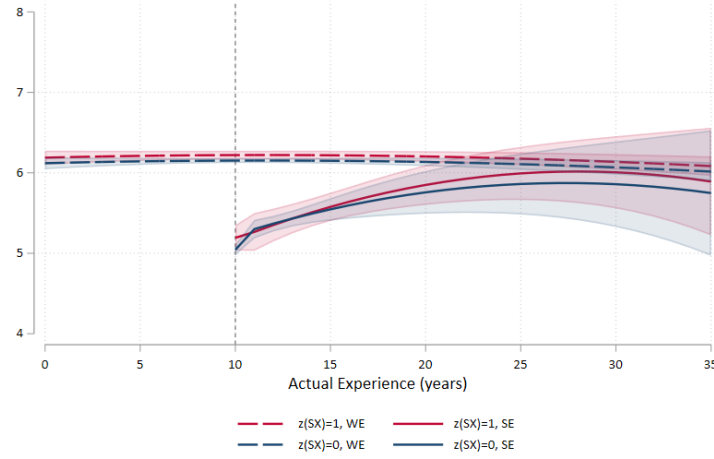
Note: This table reports coefficient estimates for the baseline earnings specification in equation (C2), estimated separately for workers who ever enter basic (columns 1–2) and advanced (columns 3–4) self-employment. Columns 1 and 3 are unweighted; columns 2 and 4 weight observations by the inverse of the number of observations within each job spell to address survivorship bias. The dependent variable is log weekly earnings in 2010 USD. Standard errors clustered at the individual level are in parentheses. \*p < .10; \*\*p < .05; \*\*\*p < .01.

Figure C1: Predicted Log Weekly Earnings for Basic Self-Employment

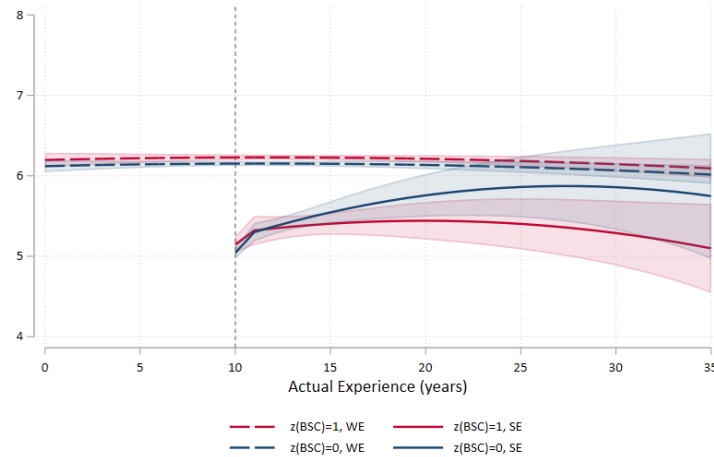
(a) by Business Skills (BX)



(b) by Specialized Skills (SX)



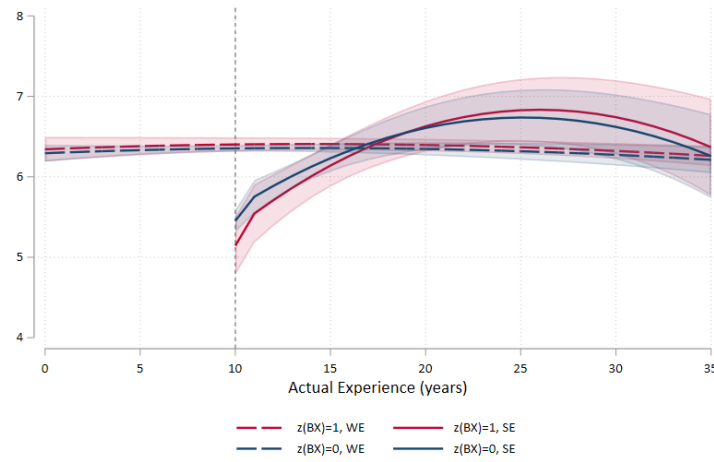
(c) by Skill Balance (BSC)



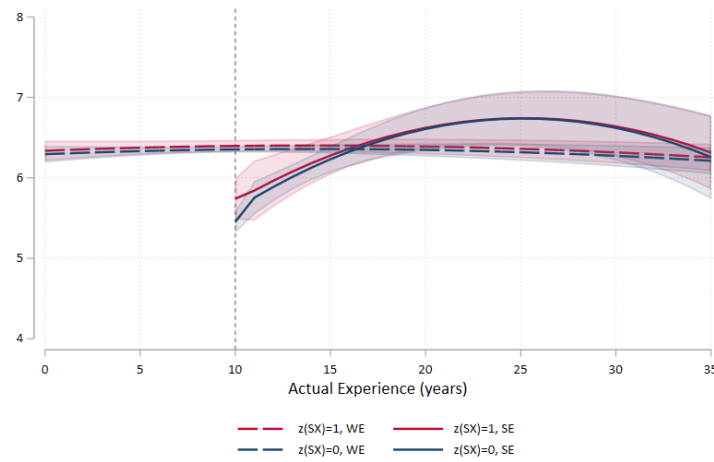
Note: This figure plots predicted log weekly earnings as a function of actual experience for workers who ever enter basic self-employment, separately for wage employment (dashed) and self-employment (solid), and for the indicated skill measure at the mean ( $z=0$ , blue) and one standard deviation above the mean ( $z=1$ , red). Predictions come from the individual fixed-effects specification in equation (C1). Self-employment entry occurs at 10 years of actual experience (vertical dashed line); other covariates are held at observed values. Shaded bands are 95 percent confidence intervals.

Figure C2: Predicted Log Weekly Earnings for Advanced Self-Employment

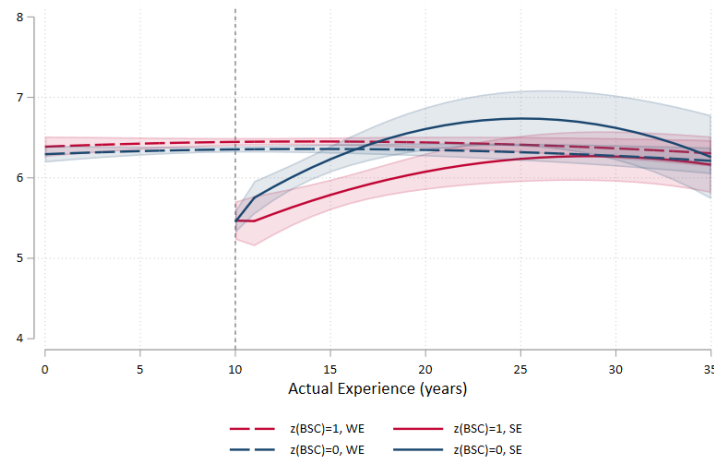
(a) by Business Skills (BX)



(b) by Specialized Skills (SX)



(c) by Skill Balance (BSC)



Note: This figure plots predicted log weekly earnings as a function of actual experience for workers who ever enter advanced self-employment, separately for wage employment (dashed) and self-employment (solid), and for the indicated skill measure at the mean ( $z=0$ , blue) and one standard deviation above the mean ( $z=1$ , red). Predictions come from the individual fixed-effects specification in equation (C1). Self-employment entry occurs at 10 years of actual experience (vertical dashed line); other covariates are held at observed values. Shaded bands are 95 percent confidence intervals.